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Ishii et al.

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(54) **IMAGE FORMING APPARATUS, FIXING UNIT, AND IMAGE FORMING METHOD WITH IMPROVED HEATING MECHANISM**

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G03G 15/20 (2006.01)

(52) **U.S. Cl.** **399/330**; 399/67; 399/338

(58) **Field of Classification Search** 399/336, 399/122, 67-70, 320, 328-330, 335, 338; 219/216, 244

See application file for complete search history.

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Primary Examiner — David Porta

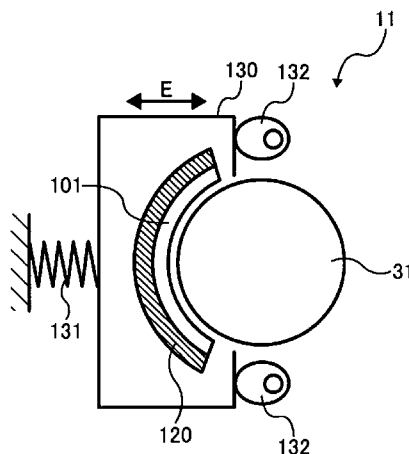
Assistant Examiner — Benjamin Schmitt

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(57) **ABSTRACT**

An image forming apparatus includes an image forming mechanism to form a toner image on a recording medium according to image data and a fixing mechanism to fix the toner image on the recording medium. The fixing mechanism includes a fixing member and an external heater. The fixing member applies heat to the recording medium having the toner image. The external heater heats the fixing member and is formed in a shape corresponding to a surface of the fixing member. The external heater is disposed as if to engage and yet be spaced apart from the fixing member by a distance which is variable according to movement of the external heater.

24 Claims, 11 Drawing Sheets



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FIG. 1

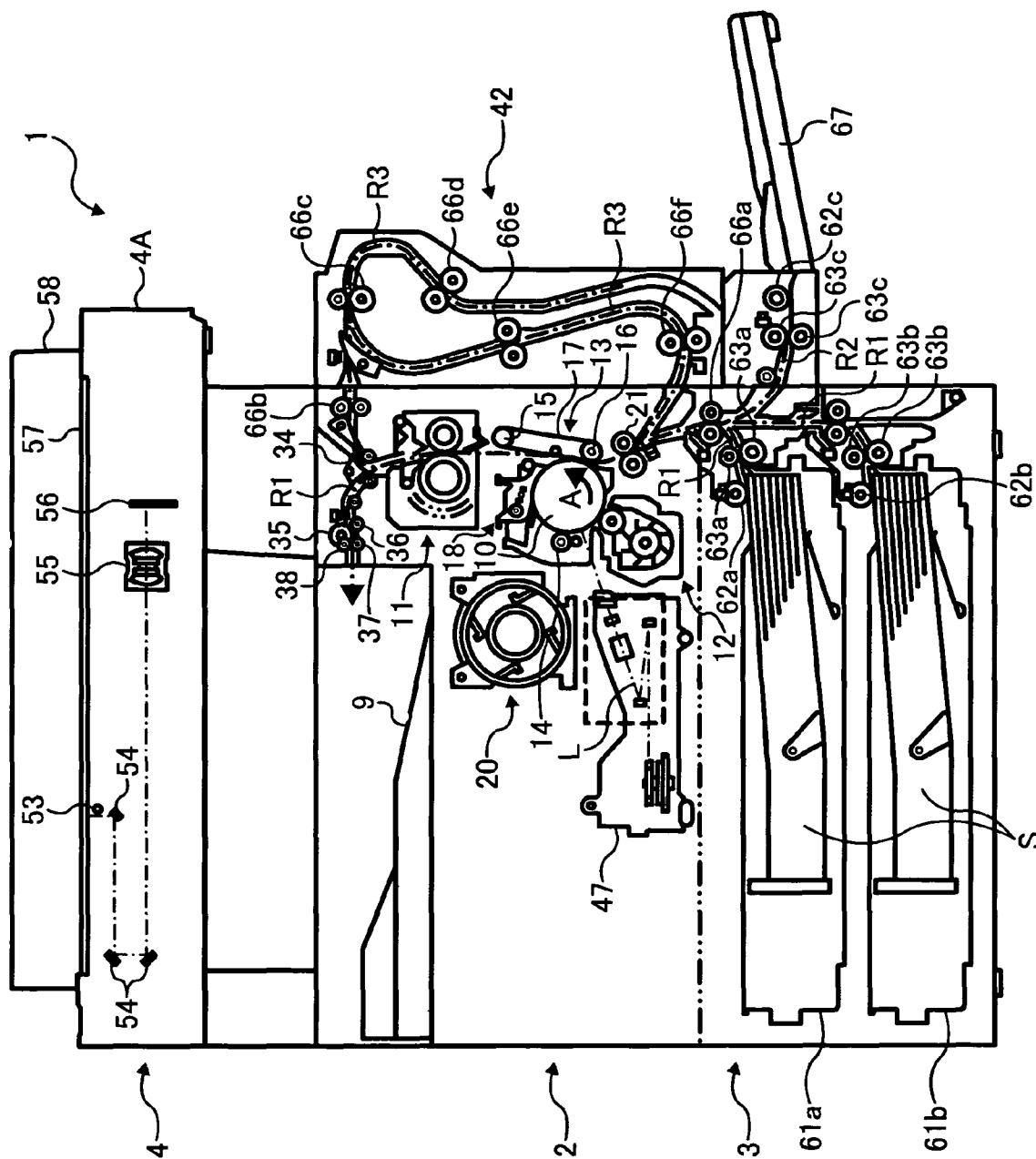


FIG. 2

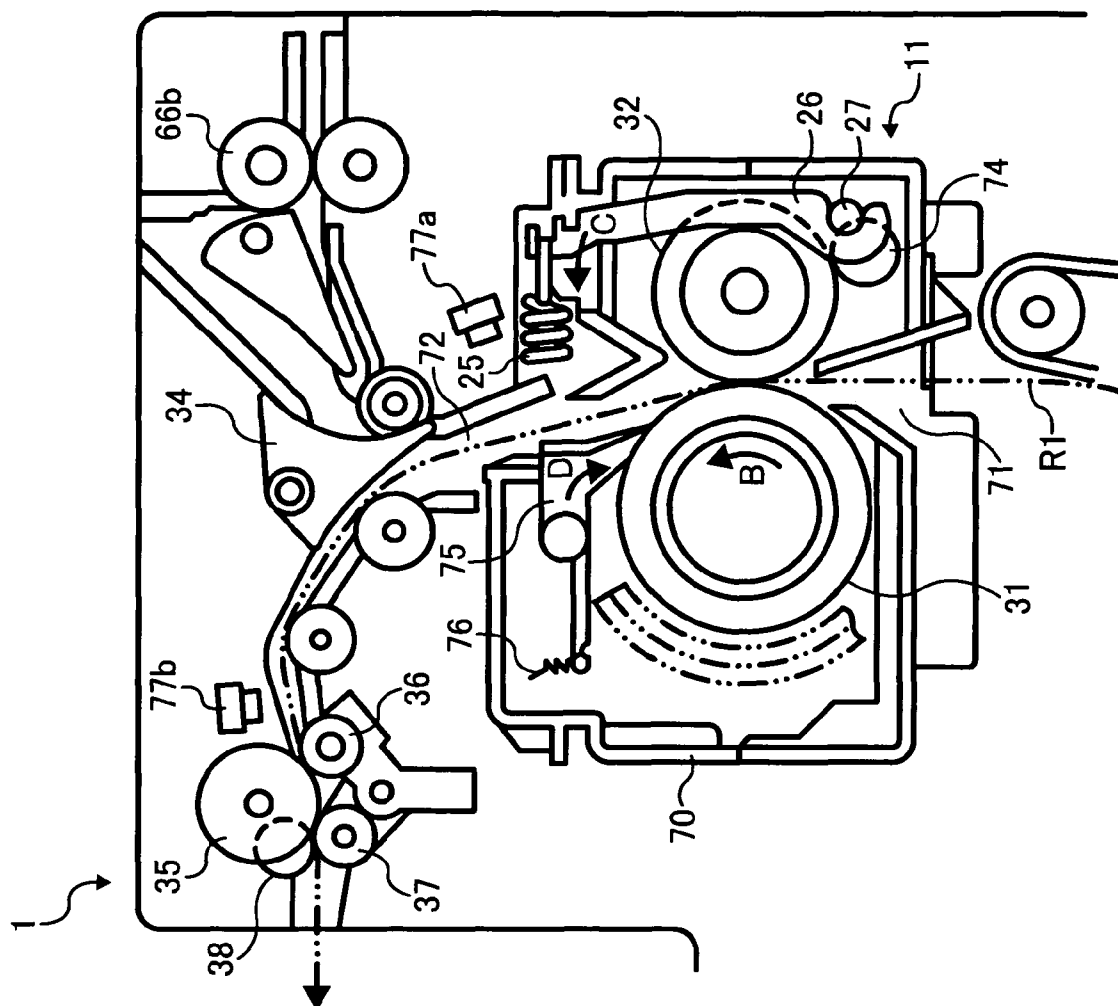


FIG. 3

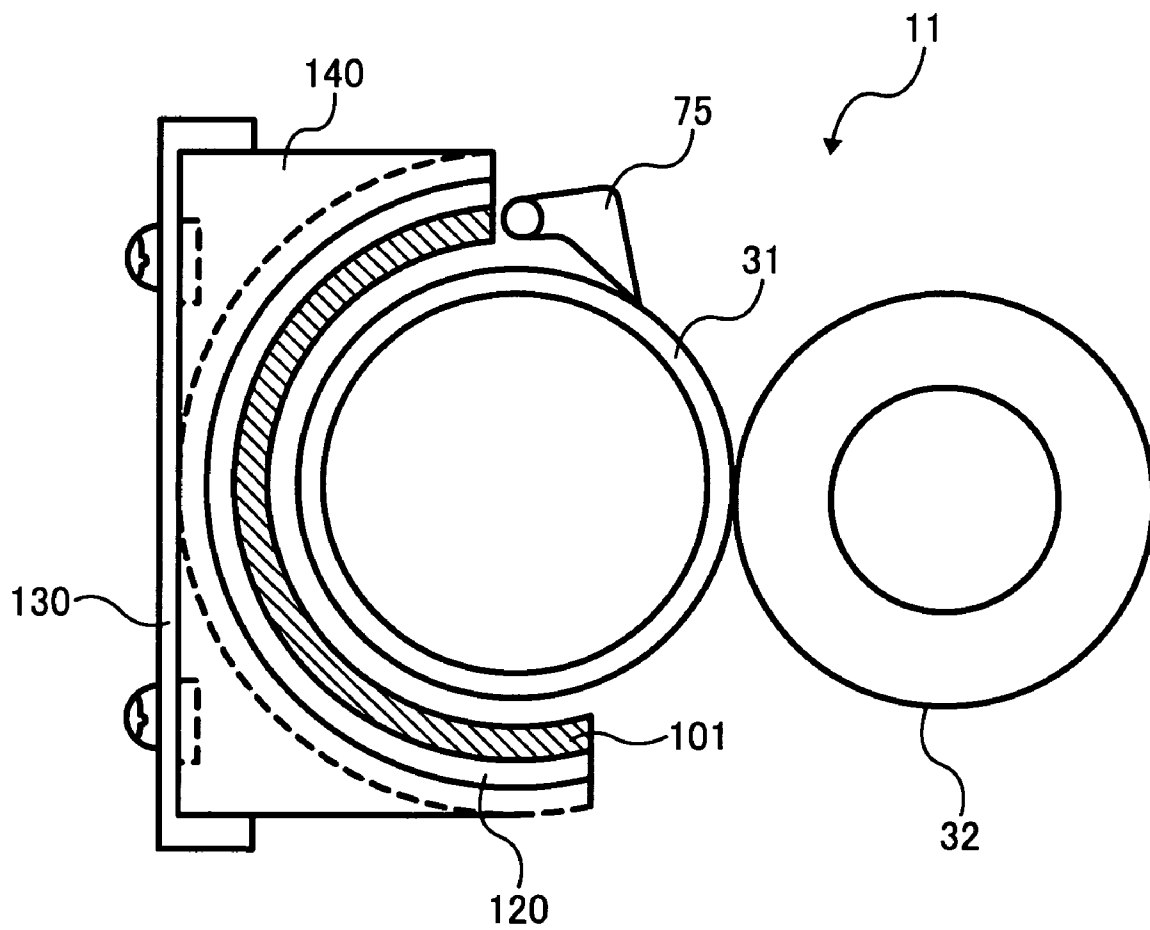


FIG. 4A

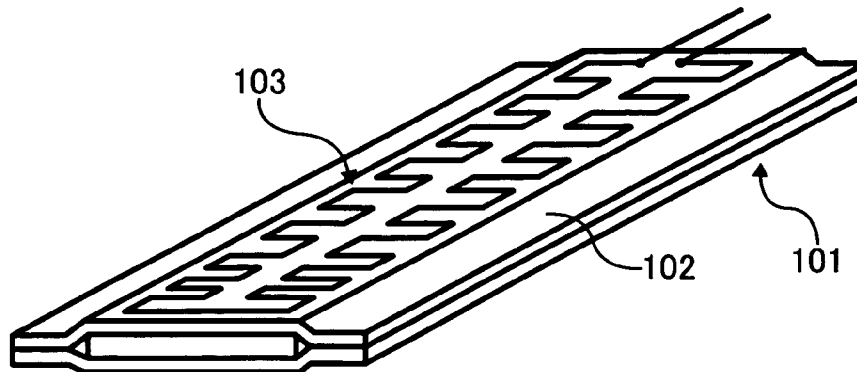


FIG. 4B

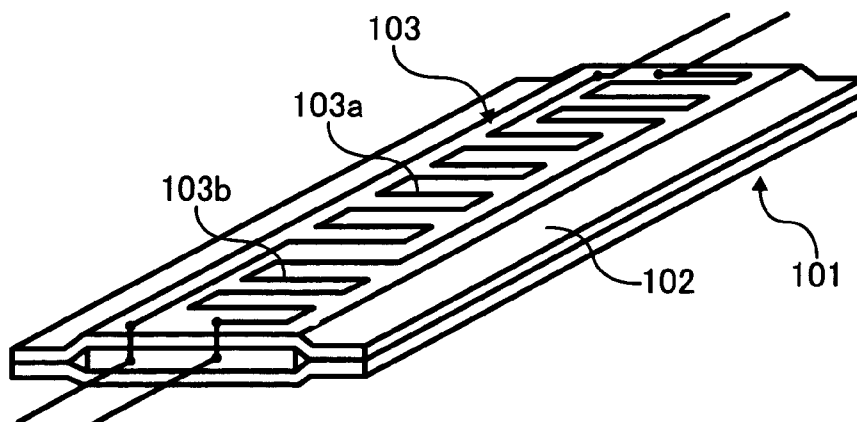


FIG. 5

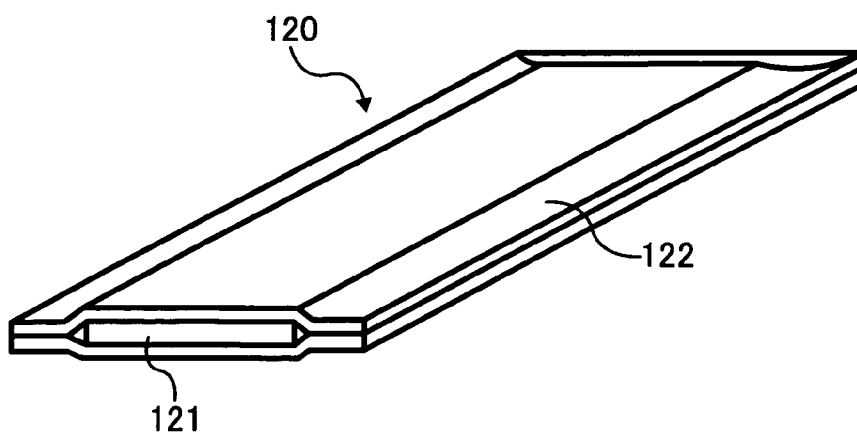


FIG. 6A

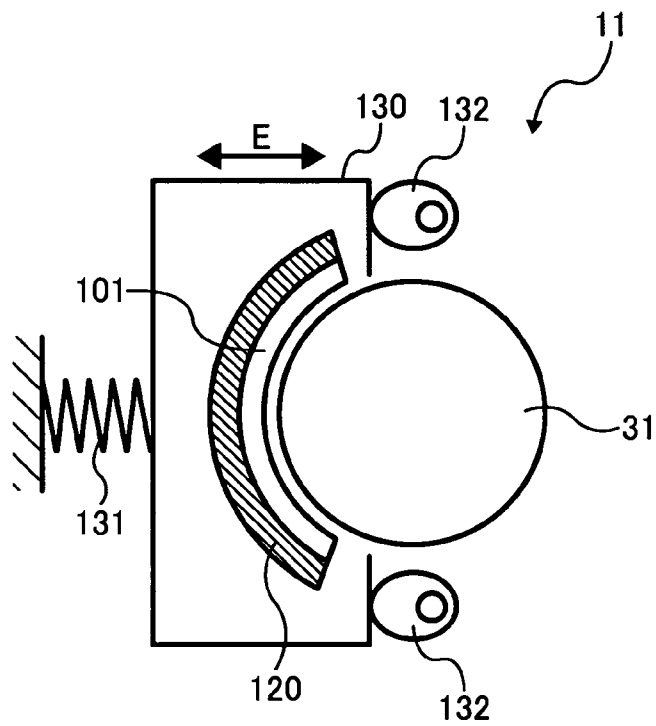


FIG. 6B

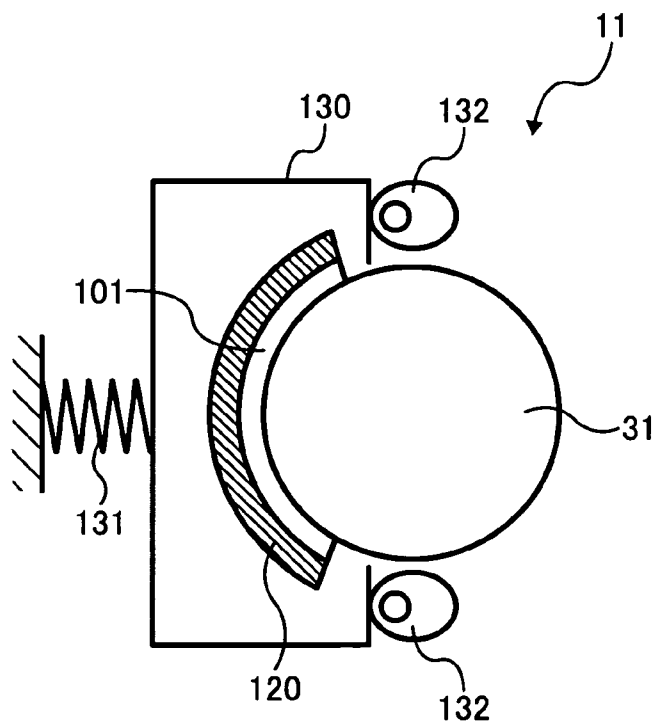


FIG. 7A

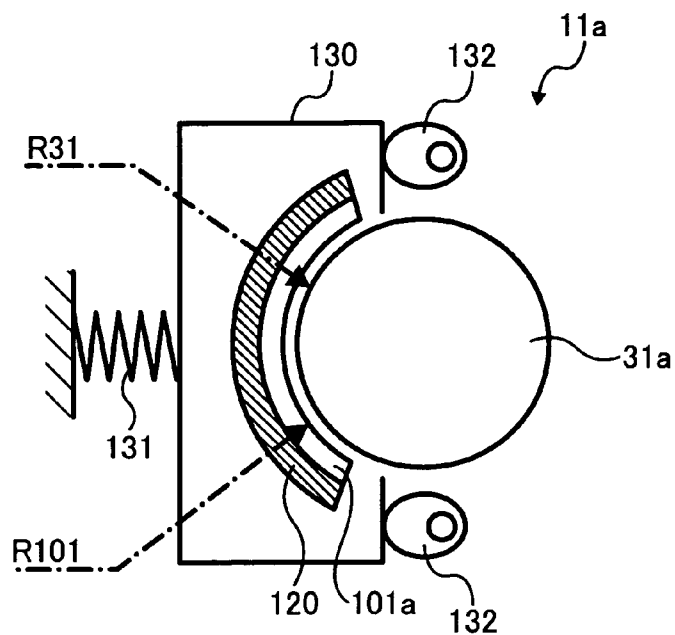


FIG. 7B

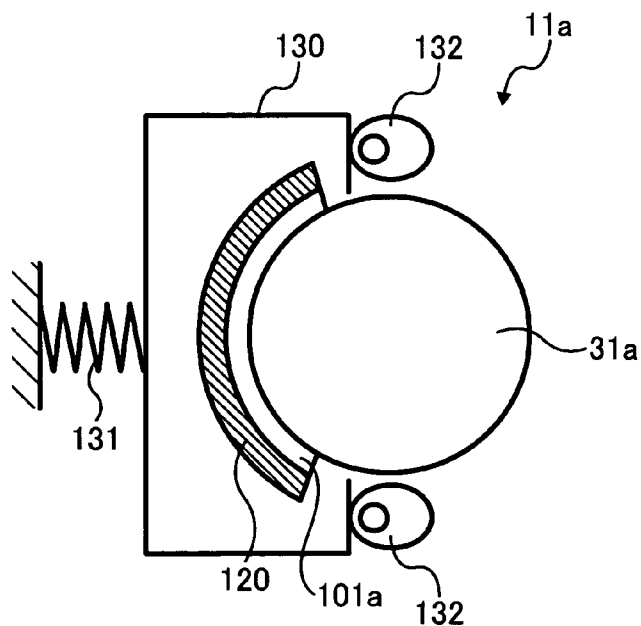
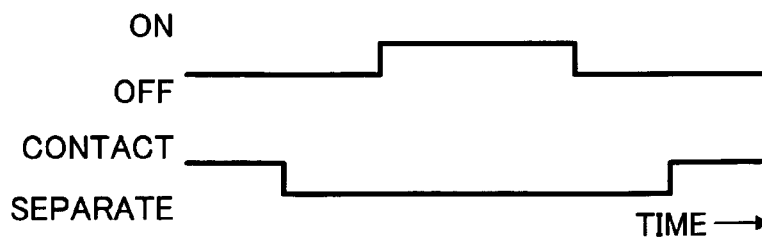


FIG. 8

ROTATE HEATING
ROLLER

CONTACT AND
SEPARATE EXTERNAL
HEATER FROM
HEATING ROLLER



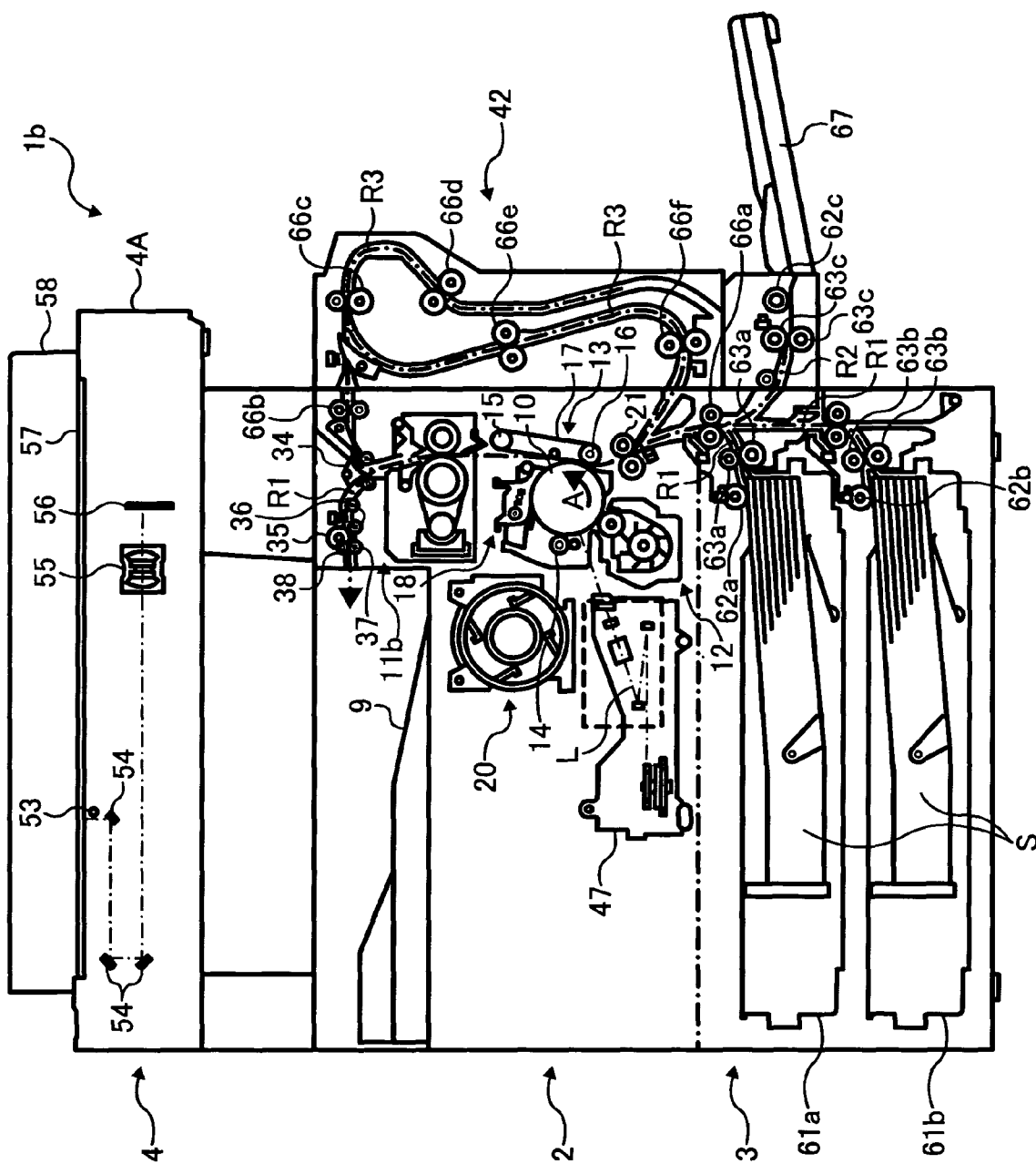


FIG. 9

FIG. 10

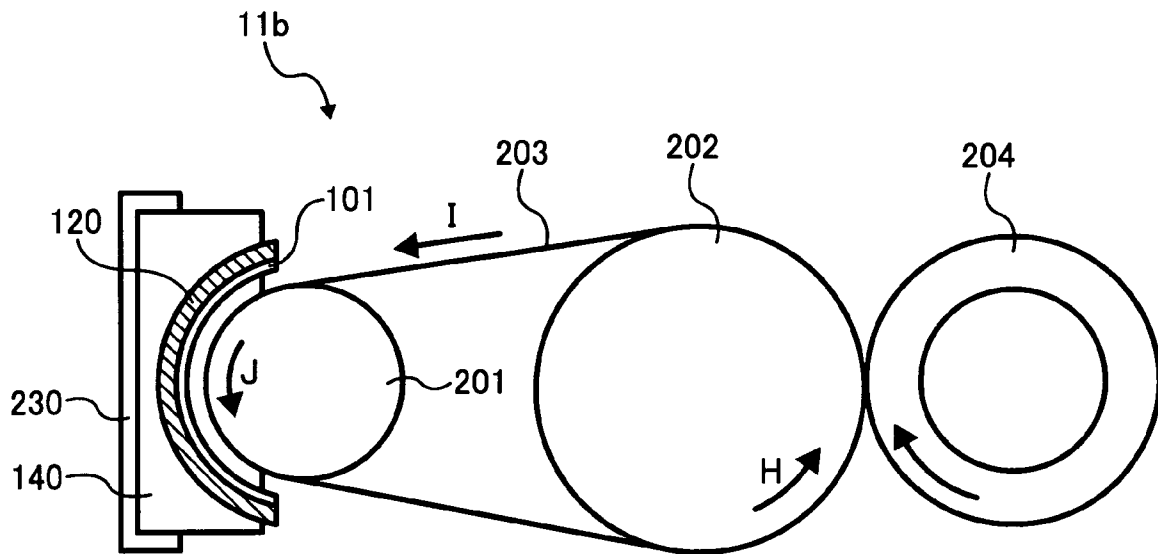


FIG. 11

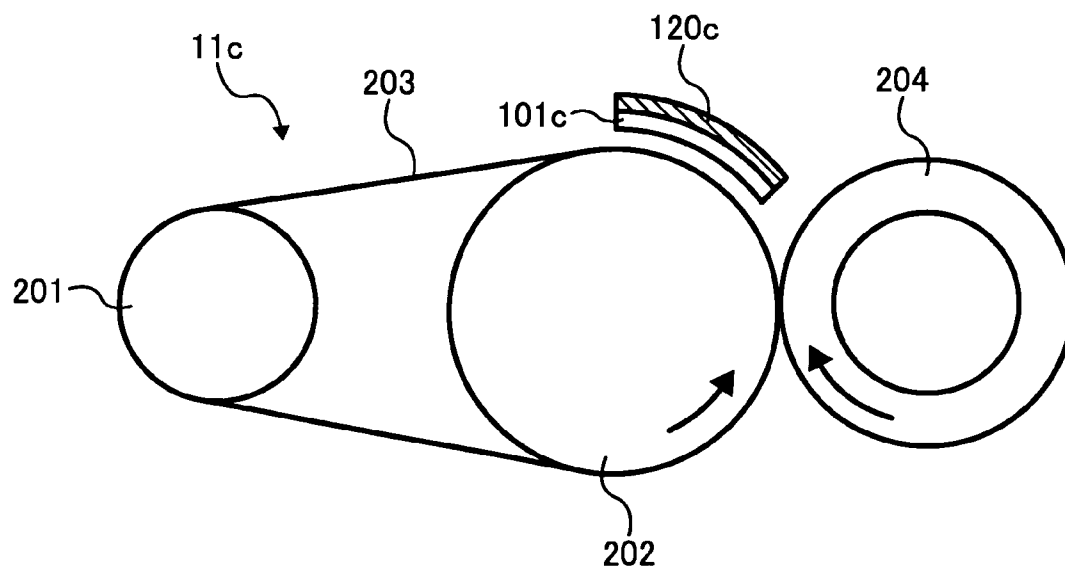


FIG. 12

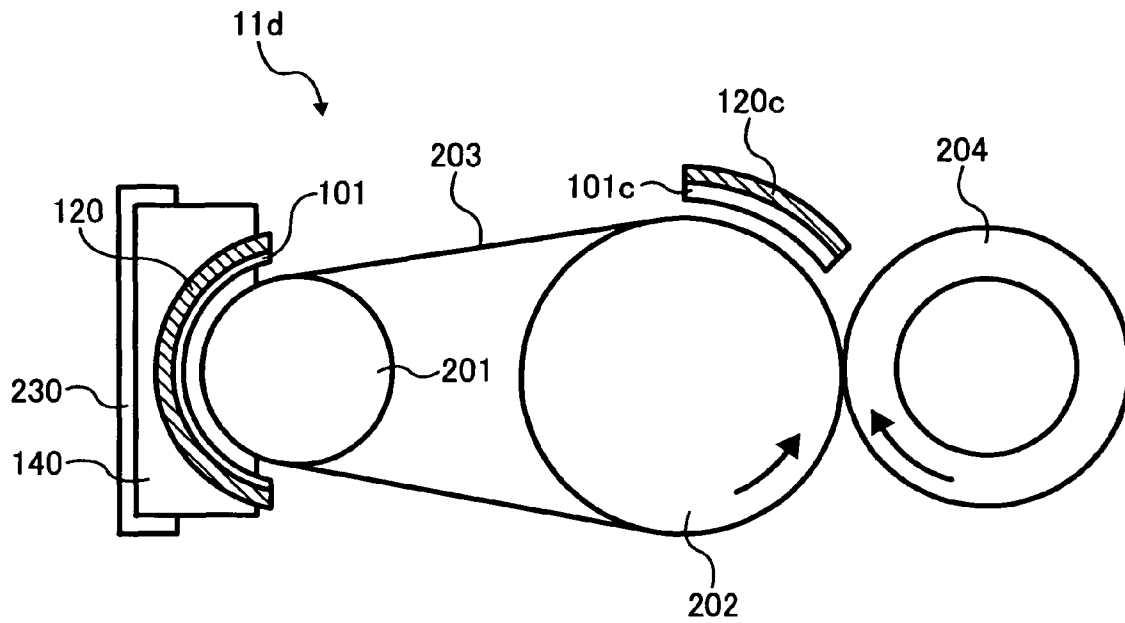


FIG. 13

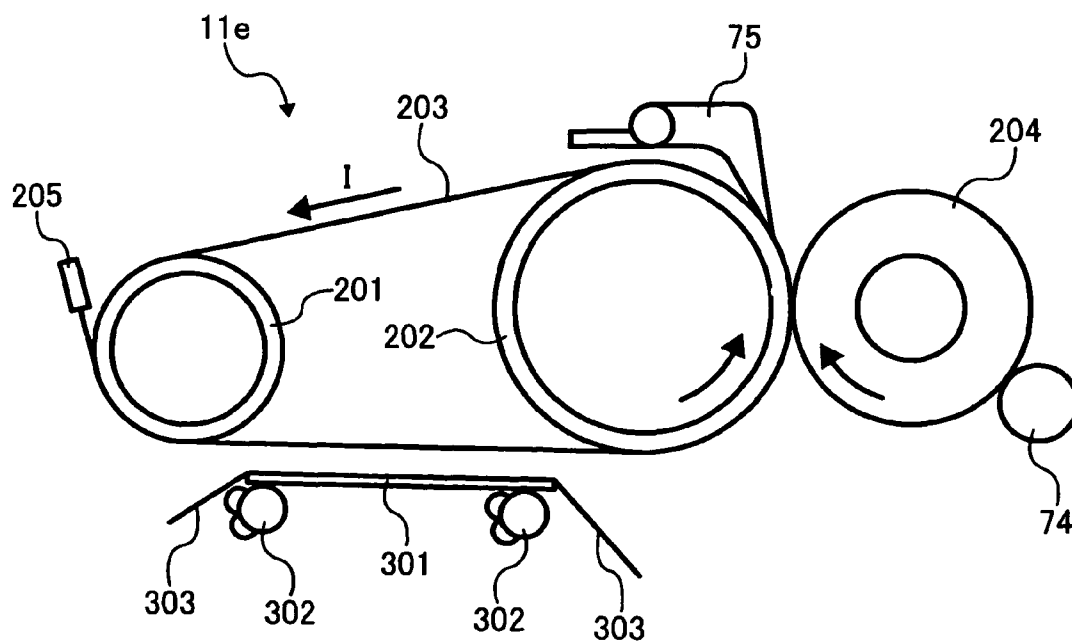


FIG. 14A

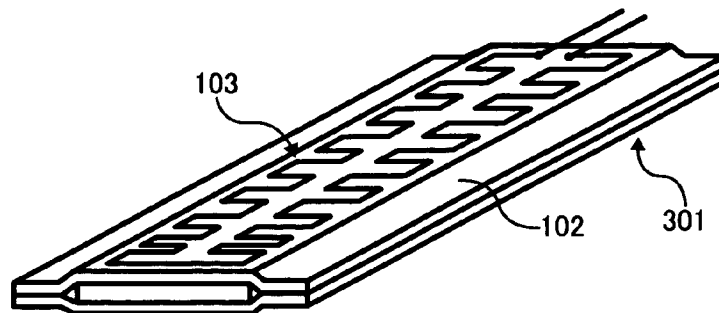


FIG. 14B

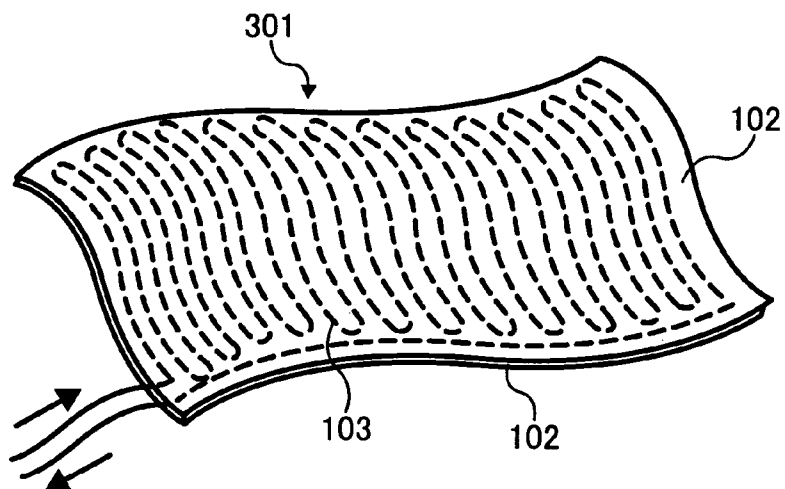


FIG. 15

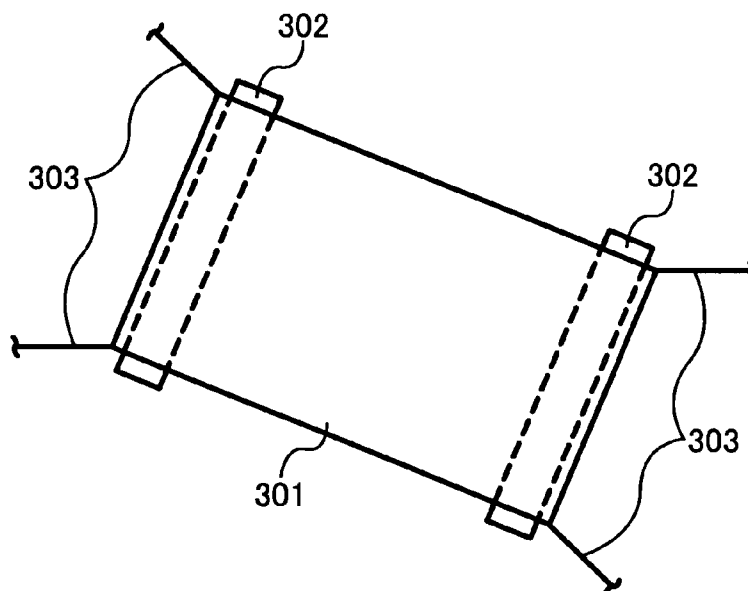


FIG. 16A

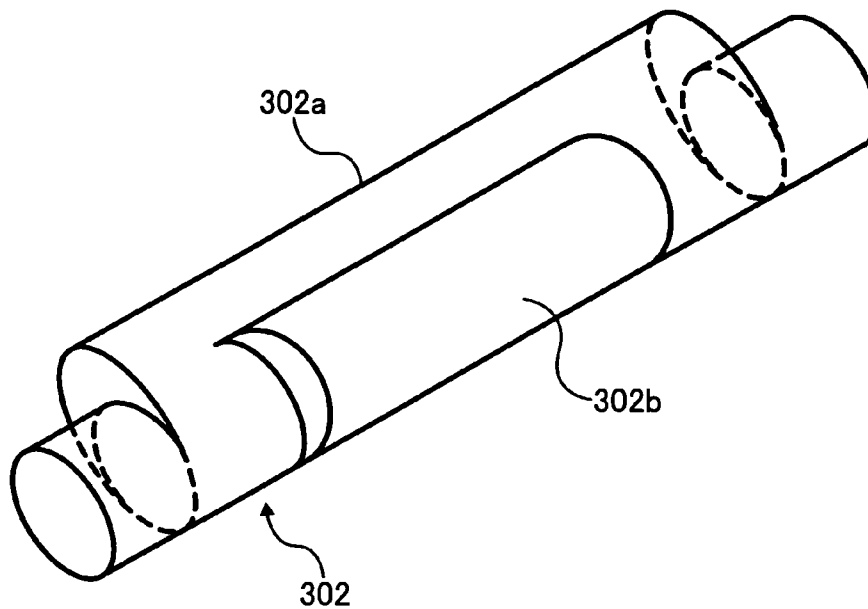


FIG. 16B

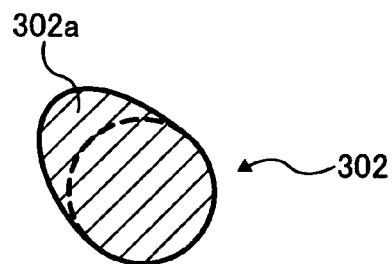
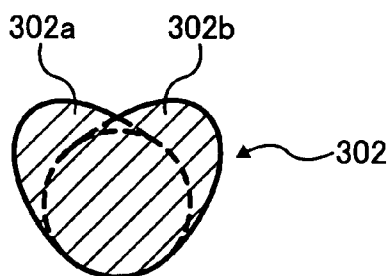


FIG. 16C



1

IMAGE FORMING APPARATUS, FIXING UNIT, AND IMAGE FORMING METHOD WITH IMPROVED HEATING MECHANISM

PRIORITY STATEMENT

This application claims the priority of Japanese Patent Application No. 2005-207838, filed on Jul. 15, 2005, the disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND

1. Technical Field

Example embodiments of the present invention generally relate to an image forming apparatus, a fixing unit, and/or an image forming method with an improved heating mechanism, e.g., for heating a fixing member to fix a toner image on a recording medium for example.

2. Description of Background Art

A background electrophotographic image forming apparatus, such as a copying machine, a printer, a facsimile machine, or a multifunction printer including copying, printing, and facsimile functions, generally forms an electrostatic latent image on a photoconductor according to image data. The electrostatic latent image is visualized with toner to form a toner image on the photoconductor. The toner image is transferred onto a sheet and the sheet having the toner image is conveyed to a fixing unit in which heat and pressure are applied to the sheet to fix the toner image on the sheet.

In such a fixing unit, a fixing member for applying heat to the sheet having the toner image needs to be heated to a fixing temperature in a shorter time period so as to enhance user-friendliness. Further, such fixing unit is desired to consume less power to save energy.

To cope with such needs, in an example background fixing unit, a fixing member is configured to have a thinner thickness (e.g., about 1 mm or less) to improve temperature response so as to be heated in a shorter time period. Power is supplied to a heater for heating the fixing member when an image forming operation starts while it is not supplied in a standby mode so as to save energy.

However, the fixing member having the thinner thickness has a small heat capacity and heat may be easily transferred to a sheet contacting the fixing member for fixing, resulting in an uneven temperature of the fixing member. To address this problem, another example background fixing unit is provided with a heat insulator for covering the fixing member to reduce heat radiation from the fixing member. However, the heat insulator may obstruct a quick heating of the fixing member.

To heat the fixing member in a shorter time period, yet another example background fixing unit is provided with an external heater disposed outside the fixing member and/or a heater disposed inside the fixing member to heat the fixing member. The external heater continuously contacts the fixing member or contacts and separates from the fixing member. However, the external heater contacting the fixing member may damage the fixing member.

SUMMARY

At least one embodiment of the present invention provides an image forming apparatus that includes an image forming mechanism to form a toner image on a recording medium according to image data and a fixing mechanism to fix the toner image on the recording medium. The fixing mechanism includes a fixing member and an external heater. The fixing

2

member applies heat to the recording medium having the toner image. The external heater heats the fixing member and is formed in a shape corresponding to a surface of the fixing member. The external heater is disposed as if to engage and yet be spaced apart from the fixing member by a distance which is variable according to movement of the external heater.

At least one embodiment of the present invention provides a fixing unit for fixing a toner image on a recording medium. Such a fixing unit includes a fixing member and an external heater. The fixing member applies heat to the recording medium having the toner image. The external heater heats the fixing member and is formed in a shape corresponding to a surface of the fixing member. The external heater is disposed as if to engage and yet be spaced apart from the fixing member by a distance which is variable according to movement of the external heater.

At least one embodiment of the present invention provides an image forming method that includes forming a toner image on a recording medium according to image data and fixing the toner image on the recording medium. The fixing step includes sub-steps of disposing an external heater (formed in a shape corresponding to a surface of a fixing member) as if to engage and yet be spaced apart from the fixing member, moving the external heater to be a desired distance apart from the fixing member, heating the fixing member with the external heater, and applying heat to the recording medium having the toner image.

Additional features and advantages of the present invention will be more fully apparent from the following detailed description of example embodiments, the accompanying drawings and the associated claims.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete appreciation of the present invention and the many attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description of example embodiments when considered in connection with the accompanying drawings, wherein:

FIG. 1 is a schematic view of an image forming apparatus according to an example embodiment of the present invention;

FIG. 2 is a more detailed schematic view (according to an example embodiment of the present invention) of a fixing unit and peripheral elements of the image forming apparatus shown in FIG. 1;

FIG. 3 is a more detailed schematic view (according to an example embodiment of the present invention) of the fixing unit shown in FIG. 2;

FIG. 4A is a perspective view (according to an example embodiment of the present invention) of an external heater for the fixing unit shown in FIG. 3;

FIG. 4B is a perspective view (according to an example embodiment of the present invention) of another external heater for the fixing unit shown in FIG. 3;

FIG. 5 is a perspective view (according to an example embodiment of the present invention) of a heat insulator of the fixing unit shown in FIG. 3;

FIG. 6A is a schematic view (according to an example embodiment of the present invention) of an external heater separated from a heating roller of the fixing unit shown in FIG. 3;

FIG. 6B is a schematic view (according to an example embodiment of the present invention) of the external heater contacting the heating roller shown in FIG. 6A;

3

FIG. 7A is a schematic view of an external heater separated from a heating roller of a fixing unit according to an example embodiment of the present invention;

FIG. 7B is a schematic view of the external heater contacting the heating roller shown in FIG. 7A;

FIG. 8 is a timing chart of operations of an external heater and a heating roller of a fixing unit according to an example embodiment of the present invention;

FIG. 9 is a schematic view of an image forming apparatus according to an example embodiment of the present invention;

FIG. 10 is a more detailed schematic view (according to an example embodiment of the present invention) of a fixing unit of the image forming apparatus shown in FIG. 9;

FIG. 11 is a more detailed schematic view of a fixing unit according to an example embodiment of the present invention;

FIG. 12 is a more detailed schematic view of a fixing unit according to an example embodiment of the present invention;

FIG. 13 is a schematic view of a fixing unit according to an example embodiment of the present invention;

FIG. 14A is a perspective view (according to an example embodiment of the present invention) of an external heater for the fixing unit shown in FIG. 13;

FIG. 14B is a perspective view (according to an example embodiment of the present invention) of another external heater for the fixing unit shown in FIG. 13;

FIG. 15 is a schematic view of the external heater, cam shafts, and tension belts of the fixing unit shown in FIG. 13;

FIG. 16A is a more detailed perspective view (according to an example embodiment of the present invention) of the cam shaft shown in FIG. 15;

FIG. 16B is a sectional view of the cam shaft shown in FIG. 16A; and

FIG. 16C is another sectional view of the cam shaft shown in FIG. 16A.

The accompanying drawings are intended to depict example embodiments of the present invention and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

It will be understood that if an element or layer is referred to as being “on,” “against,” “connected to” or “coupled to” another element or layer, then it can be directly on, against, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, if an element is referred to as being “directly on,” “directly connected to” or “directly coupled to” another element or layer, then there are no intervening elements or layers present. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented

4

“above” the other elements or features. Thus, term such as “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

Although the terms first, second, etc. may be used herein to describe various elements, components, regions, layers and/or sections, it should be understood that these elements, components, regions, layers and/or sections should not be limited by these terms. These terms are used only to distinguish one element, component, region, layer or section from another region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present invention.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present invention. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes” and/or “including,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

In describing example embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this patent specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that operate in a similar manner.

Referring now to the drawings, wherein like reference numerals designate identical or corresponding parts throughout the several views, particularly to FIG. 1, an image forming apparatus 1 according to an example embodiment of the present invention is explained.

As illustrated in FIG. 1, the image forming apparatus 1 includes a reader 4, a paper tray unit 3, a body 2, a conveyance path R1, and/or a duplex unit 42.

The image forming apparatus 1 includes a copying machine, a printer, a facsimile machine, a multifunction printer including copying, printing, and facsimile functions, or the like which forms an image on a recording medium (e.g., a sheet and an OHP (overhead projector) transparency) in an electrophotographic method. According to this example embodiment, the image forming apparatus 1 functions as a copying machine, e.g., a monochrome copying machine, for scanning an image on an original to create image data and forming a monochrome image on a sheet according to the image data.

The reader 4 is disposed on the body 2 and is configured to scan an image on an original to create image data and to send the image data to the body 2. Types of recording media other than, or in addition to, paper can be used. The paper tray unit 3 is disposed under the body 2 and is configured to load sheets S (i.e., the recording medium including the sheet and the OHP transparency) and to feed the sheets S one by one to the body 2. The body 2 is disposed on the paper tray unit 3 and is configured to perform image processing to form an image on the sheet S sent from the paper tray unit 3 according to the image data created by the reader 4. The conveyance path R1 is provided in the paper tray unit 3 and the body 2 to convey the sheet S in the paper tray unit 3 and the body 2. The conveyance path R1 mostly extends in a substantially vertical

5

direction. The duplex unit 42 is configured to reverse the sheet S sent from the body 2 and to send back the sheet S to the body 2 for duplex copying.

The reader 4 includes an exposure glass cover 58 and/or a body 4A. The body 4A includes an exposure glass 57, a first traveler 53, second travelers 54, a lens 55, and/or a reading sensor 56.

The exposure glass 57 forms a part of a top of the body 4A. The exposure glass cover 58 has a larger size than the exposure glass 57 and is hinged on the top of the body 4A so as to be moved from a lowered position to a lifted position or from the lifted position to the lowered position. A user lifts the exposure glass cover 58 from the lowered position to the lifted position to place an original to be scanned onto the exposure glass 57. The user then lowers the exposure glass cover 58 so that the exposure glass cover 58 presses the original placed on the exposure glass 57. Thus, the original contacts the exposure glass 57 and is set flat on the exposure glass 57 in a state that an image on the original may be properly scanned.

When the user presses a start button on a control panel (not shown) of the image forming apparatus 1, the reader 4 starts scanning the image on the original. Specifically, a driving mechanism (not shown) moves the first traveler 53 and the second travelers 54. A light source of the first traveler 53 emits light onto the original. Mirrors of the second travelers 54 deflect the light reflected by the original toward the lens 55. The lens 55 irradiates the deflected light into the reading sensor 56. The reading sensor 56 converts the light into an electric signal to create image data.

The paper tray unit 3 includes paper trays 61a and 61b, a bypass tray 67, feeding rollers 62a, 62b, and 62c, separating roller pairs 63a, 63b, and 63c, the conveyance path R1, a conveyance path R2, and/or a feeding roller 66a.

The paper tray 61a, the paper tray 61b, and the bypass tray 67 load sheets S. The paper trays 61a and 61b are layered in the paper tray unit 3. The paper trays 61a and 61b may load sheets S different in size and orientation from each other. The paper tray 61a or 61b is automatically selected according to the size and orientation of the original placed on the exposure glass 57 or manually selected by the user according to the size and orientation the user has specified on the control panel. When the user presses the start button on the control panel, the feeding roller 62a or 62b rotates to feed an uppermost sheet S of the sheets S loaded on the paper tray 61a or 61b. The separating roller pair 63a or 63b separates the uppermost sheet S from other sheet S when a plurality of sheets S is fed by the feeding roller 62a or 62b and feeds only the uppermost sheet S toward the conveyance path R1. The feeding roller 66a feeds the sheet S conveyed through the conveyance path R1 up toward the body 2.

The bypass tray 67 may load sheets S including thick paper, a postcard, and an OHP transparency. When the sheets S are placed on the bypass tray 67, the bypass tray 67 is automatically selected. When the user presses the start button on the control panel, the feeding roller 62c rotates to feed an uppermost sheet S of the sheets S loaded on the bypass tray 67. The separating roller pair 63c separates the uppermost sheet S from other sheet S when a plurality of sheets S is fed by the feeding roller 62c and feeds only the uppermost sheet S toward the conveyance path R2. The feeding roller 66a feeds the sheet S conveyed through the conveyance path R2 up toward the body 2.

The body 2 includes a registration roller 21, a photoconductor 10, a charger 14, an exposure unit 47, a development unit 12, a transferor 13, a cleaning unit 18, a toner container 20, a fixing unit 11, a switching nail 34, feeding rollers 35, 36, 37, and 38, an output tray 9, and/or a feeding roller 66b.

6

The sheet S fed by the feeding roller 66a stops when contacting the registration roller 21.

The photoconductor 10 is formed in a drum shape (e.g., a cylindrical shape) and is rotatably supported by a shaft extending in a horizontal direction so as to rotate in a rotating direction A. A driver (not shown) including a motor drives the photoconductor 10 to rotate in the rotating direction A at a constant speed.

The charger 14, the development unit 12, the transferor 13, the cleaning unit 18, and a discharger (not shown) are disposed around the photoconductor 10 in this order in the rotating direction A. The exposure unit 47 is disposed obliquely downward of the photoconductor 10. Charging, exposing, developing, transferring, cleaning, and discharging positions are formed in this order in the rotating direction A on a surface of the photoconductor 10 in accordance with the charger 14, the exposure unit 47, the development unit 12, the transferor 13, the cleaning unit 18, and the discharger disposed as described above.

The charger 14 uniformly charges the surface of the photoconductor 10 at the charging position. The exposure unit 47 emits light L (e.g., laser beam) onto the exposing position to form an electrostatic latent image on the surface of the photoconductor 10 according to the image data. The development unit 12 includes a toner brush (not shown) carrying toner particles and visualizes the electrostatic latent image at the developing position with the toner particles of the toner brush to form a toner image.

The transferor 13 includes a support roller 15, a support roller 16, and/or a transfer belt 17. The support rollers 15 and 16 oppose each other in a state that a distance is provided between the support rollers 15 and 16 in a substantially vertical direction. The transfer belt 17 is looped over the support rollers 15 and 16. The registration roller 21 starts rotating to feed the sheet S to the transferor 13 at a timing when the toner image formed on the surface of the photoconductor 10 is properly transferred onto the sheet S. As a result, the toner image is transferred onto the sheet S at the transferring position while the sheet S is conveyed on the transfer belt 17. The transfer belt 17 conveys the sheet S having the toner image up toward the fixing unit 11. Namely, a portion of the support roller 16, over which the transfer belt 17 is looped, presses the photoconductor 10 via the transfer belt 17 to form the transferring position between the surface of the photoconductor 10 and an outer circumferential surface of the transfer belt 17. The support roller 15 is disposed near the fixing unit 11.

The cleaning unit 18 includes a blade (not shown), a brush (not shown), or both of the blade and the brush for removing contaminants including residual toner particles not transferred and remaining on the surface of the photoconductor 10. The blade includes an edge for pressingly contacting the surface of the photoconductor 10 at the cleaning position. The brush contacts the surface of the photoconductor 10 at the cleaning position and is driven to rotate in accordance with rotation of the photoconductor 10.

The discharger includes a lamp for emitting light. The lamp emits light onto the surface of the photoconductor 10 at the discharging position to discharge the surface of the photoconductor 10 so that the surface potential of the photoconductor 10 is restored to an initial level.

The toner container 20 contains unused toner particles and includes a toner bottle. A toner supplying path (not shown) is provided between the toner container 20 and the development unit 12. The unused toner particles are supplied from the toner container 20 to the development unit 12 when the development unit 12 has an insufficient quantity of toner particles

7

after using the toner particles for visualizing the electrostatic latent image formed on the surface of the photoconductor 10.

As described above, when the user presses the start button on the control panel, the photoconductor 10 starts rotating so that the toner image is formed on the surface of the photoconductor 10 according to the image data. Namely, while the photoconductor 10 rotates, a portion on the surface of the photoconductor 10 passes the charging, exposing, developing, transferring, cleaning, and discharging positions formed on the surface of the photoconductor 10 in accordance with the charger 14, the exposure unit 47, the development unit 12, the transferor 13, the cleaning unit 18, and the discharger. An image forming operation is performed in a cycle including charging, exposing, developing, transferring, cleaning, and discharging. The cycle is repeated to form another toner image within a suitable distance along the outer circumferential surface of the photoconductor 10 in the rotating direction A according to the size of the toner image to be formed.

In the fixing unit 11, heat and pressure are applied to the sheet S to fix the toner image on the sheet S. The sheet S having the fixed toner image is fed toward the switching nail 34. The switching nail 34 guides the sheet S through the conveyance path R1 toward the feeding rollers 35, 36, 37, and 38. The feeding rollers 35, 36, 37, and 38 feed the sheet S onto the output tray 9 disposed in an upper portion of the body 2.

When the user has selected duplex copying, the switching nail 34 guides the sheet S toward the feeding roller 66b. The feeding roller 66b feeds the sheet S toward the duplex unit 42.

The duplex unit 42 includes the feeding rollers 66c, 66d, 66e, and 66f and/or a reverse conveyance path R3. The feeding roller 66c feeds the sheet S through the reverse conveyance path R3 toward the feeding roller 66d. Then, the feeding rollers 66d and 66c feed the sheet S back toward the feeding roller 66e. The feeding roller 66e feeds the sheet S through the reverse conveyance path R3 toward the feeding roller 66f. The feeding roller 66f feeds the sheet S toward the registration roller 21. The registration roller 21 feeds the sheet S toward the transferor 13. In the transferor 13, a toner image is transferred onto the other side of the sheet S and the sheet S having the toner image is fed toward the fixing unit 11. In the fixing unit 11, the toner image on the other side of the sheet S is fixed and the sheet S having the fixed toner image is fed toward the feeding rollers 35, 36, 37, and 38. The feeding rollers 35, 34, 37, and 38 feed the sheet S onto the output tray 9.

As illustrated in FIG. 2, the fixing unit 11 includes a housing 70, a heating roller 31, a pressure roller 32, a pressing lever 26, a support axis 27, a spring 25, a cleaner 74, a separating nail 75, and/or spring 76. The image forming apparatus 1 further includes paper jam sensors 77a and 77b.

The housing 70 forms an outer case covering elements of the fixing unit 11 and rotatably supports a shaft of the heating roller 31 horizontally extended. The heating roller 31 functions as a fixing member for applying heat to a sheet S having a toner image. The housing 70 also rotatably supports the pressure roller 32. The pressure roller 32 and the heating roller 31 oppose each other forms substantially horizontal direction in a state that the pressure roller 32 pressingly contacts the heating roller 31. A nip formed between the heating roller 31 and the pressure roller 32 opposing each other forms a fixing position where heat and pressure are applied to the sheet S having the toner image which is conveyed through the nip formed between the heating roller 31 and the pressure roller 32. The toner image formed on the sheet S is melted and fixed onto the sheet S.

The heating roller 31 is formed in a cylindrical shape. The housing 70 rotatably supports heating roller 31 in a state that the heating roller 31 extends in a substantially horizontal

8

direction. A driver (not shown) including a motor drives and rotates the heating roller 31 in a rotating direction B at a constant speed. Specifically, bearings (not shown) rotatably support the shaft of the heating roller 31 and an axis of rotation is fixed in the housing 70.

The pressure roller 32 is formed in a cylindrical shape and has a diameter smaller than the diameter of the heating roller 31. The pressure roller 32 is disposed to oppose the heating roller 31 in the substantially horizontal direction. The housing 70 rotatably supports the pressure roller 32 in a state that the pressure roller 32 extends in the substantially horizontal direction and is movable toward the heating roller 31. A force is applied to the pressure roller 32 to press the pressure roller 32 toward the heating roller 31.

Specifically, bearings (not shown) support the pressure roller 32 in a manner that the pressure roller 32 is movable toward the heating roller 31 within a suitable range. The pressing lever 26 presses the pressure roller 32 toward the heating roller 31.

The pressing lever 26 is formed in a shape extending for a length in a substantially vertical direction. The support axis 27 is fixed to the housing 70 and swingably supports a lower end of the pressing lever 26. An upper end of the pressing lever 26 is connected with one end of the spring 25. The other end of the spring 25 is fixed to the housing 70 in a state that the spring 25 is stretched to have a length longer than its natural length. Thus, a tension of the spring 25 applies a force for rotating the pressing lever 26 in a rotating direction C around the support axis 27. A middle portion of the pressing lever 26 pressingly contacts the pressure roller 32 to move the pressure roller 32 toward the heating roller 31. Thus, the pressure roller 32 constantly contacts the heating roller 31 with a pressure.

The housing 70 includes an entrance 71 and/or an exit 72. The entrance 71 is provided in a lower portion of the housing 70 and forms an opening formed in a slit-like shape, through which the sheet S is conveyed into the fixing unit 11 on the conveyance path R1. The exit 72 is provided in an upper portion of the housing 70 and forms an opening formed in a slit-like shape, through which the sheet S is conveyed out of the fixing unit 11 on the conveyance path R1. Thus, a path extending from the entrance 71 to the exit 72 via the nip formed between the heating roller 31 and the pressure roller 32 (i.e., a fixing position) is formed as a part of the conveyance path R1 in the housing 70.

The cleaner 74 includes a cleaning member (e.g., a cleaning roller) which contacts the pressure roller 32 and is driven by the pressure roller 32 to rotate to clean an outer circumferential surface of the pressure roller 32 after a fixing operation is performed. Namely, the cleaning member of the cleaner 74 removes contaminants including toner particles and paper dust adhered to the outer circumferential surface of the pressure roller 32 from the sheet S conveyed in the fixing unit 11 for the fixing operation.

The separating nail 75 pressingly contacts an outer circumferential surface of the heating roller 31 to prevent the sheet S from wrapping around the heating roller 31. The separating nail 75 is formed in a substantially L-like shape and includes a middle portion, a head, and a base. The housing 70 rotatably supports the middle portion. The head is disposed to contact the outer circumferential surface of the heating roller 31 at a position which is downstream of the nip formed between the heating roller 31 and the pressure roller 32 and upstream of the exit 72 in a sheet conveyance direction. The base is connected to an end of the spring 76 which is stretched. A tension of the spring 76 applies a force for rotating the separating nail 75 in a rotating direction D.

The paper jam sensors **77a** and **77b** are disposed near the conveyance path **R1** to detect a paper jam. The paper jam sensors **77a** and **77b** include an optical sensor capable of detecting a jammed sheet **S** without contacting the sheet **S**. Each of the paper jam sensors **77a** and **77b** detects an area on the conveyance path **R1**. Each of the paper jam sensors **77a** and **77b** detects the sheet **S** which is conveyed in the area on the conveyance path **R1** to locate the jammed sheet **S** when the sheet **S** stops.

A temperature sensor (not shown) is disposed near the heating roller **31**. The temperature sensor contacts or does not contact the surface of the heating roller **31** to detect a temperature of the surface of the heating roller **31**. While the image forming apparatus **1** is powered on, the temperature sensor constantly detects the temperature of the surface of the heating roller **31**. Thus, the temperature of the surface of the heating roller **31** may be controlled based on the detected temperature.

As illustrated in FIG. 3, the fixing unit **11** further includes an external heater **101**, a movable housing **130**, a resin member **140**, and/or a heat insulator **120**.

The external heater **101** is disposed along the outer circumferential surface of the heating roller **31** to form a concave-like shape with respect to the outer circumferential surface of the heating roller **31** with a distance (or gap) provided between the external heater **101** and the heating roller **31**. The external heater **101** may contact or may not contact the heating roller **31**.

Basically, the external heater **101** is deformed in a shape with which the external heater **101** is disposed along the outer circumferential surface of the heating roller **31**. Specifically, the external heater **101** deformed in the above-described shape is disposed between the movable housing **130** and the heating roller **31**. Further, the external heater **101** is selectively movable between a contact position at which the external heater **101** contacts the heating roller **31** and a non-contact position at which the external heater **101** does not contact the heating roller **31** within an area formed between the movable housing **130** and the heating roller **31** while the external heater **101** maintains the above-described shape.

The resin member **140** is attached to the movable housing **130**. The external heater **101** is fixed to the resin member **140** via the heat insulator **120**. Specifically, the movable housing **130** includes a frame of a sufficient stiffness and may be disposed inside the housing **70** forming the outer case of the fixing unit **11**. The resin member **140** includes a plastic material and is fixed to the movable housing **130**. For example, the resin member **140** is formed in a plate-like shape and a circumferential edge of the resin member **140** facing the heating roller **31** is formed in an arc-like shape which forms a concave-like shape portion with respect to the outer circumferential surface of the heating roller **31**. Thus, the external heater **101** is fixed to the circumferential edge of the resin member **140** via the heat insulator **120**. A plurality of the resin members **140** are arranged in a longitudinal direction of a shaft of the heating roller **31** in a state that a distance is provided between the adjacent resin members **140**.

The heat insulator **120** and the external heater **101** are formed in a shape with which outer circumferential surfaces of the heat insulator **120** and the external heater **101** are disposed along the circumferential edge of the resin member **140** and inner circumferential surfaces of the heat insulator **120** and the external heater **101** are disposed along the outer circumferential surface of the heating roller **31**. The heat insulator **120** and the external heater **101** are formed in the arc-like shape which forms the concave-like shape portion with respect to the outer circumferential surface of the heating

roller **31**. The external heater **101** is fixed to the resin member **140** via the heat insulator **120** in a state that the external heater **101** includes a concave-like shape portion corresponding to a convex-like shape of the outer circumferential surface of the heating roller **31** facing the external heater **101**.

As illustrated in FIG. 4A, the external heater **101** includes a heat-resistant resin sheet **102** and/or a heat generator **103**. The heat generator **103** is buried in the heat-resistant resin sheet **102** including a polyimide film having a sufficient heat resistance capable of resisting a fixing temperature. The heat-resistant resin sheet **102** is formed in a substantially rectangular, sheet-like shape having a thickness in a range of from about 0.5 mm to about 1.0 mm at least before the heat-resistant resin sheet **102** is fixed in the fixing unit **11**. The heat generator **103** includes an electrode and an external terminal crimped to the electrode. Specifically, the external terminal is connected to the electrode by crimping. The external terminal is wired to a power source (not shown) of the fixing unit **11** or the image forming apparatus **1**. Thus, power needed for the heat generator **103** to generate heat is supplied to the heat generator **103**.

The heat generator **103** may be formed in a sheet-like shape or a linear shape. The heat generator **103** having the sheet-like shape is disposed in an entire surface of the heat-resistant resin sheet **102**. The heat generators **103** having the linear shape are arranged in linear groups in the heat-resistant resin sheet **102**.

The heat generator **103** is a heater configured to have a preset resistance corresponding to the fixing temperature. The temperature of the heat generator **103** may increase up to a temperature regulated by the preset resistance. Therefore, the heat generator **103** does not need a thermostat, a thermal fuse, or a temperature controller for controlling the temperature of the heat generator **103**. Thus, electrical elements needed for controlling the temperature of the heat generator **103** may be reduced in the image forming apparatus **1**. The image forming apparatus **1** may have a simpler structure with fewer elements, providing cost reduction and improved maintenance and reliability.

FIG. 4A illustrates an example wiring of the heat generator **103** having the sheet-like shape in which heat is simultaneously generated in the entire heat generator **103**. As illustrated in FIG. 4A, the heat generator **103** includes one heat generating wire. The heat generating wire serpentine in one half of the heat-resistant resin sheet **102** in a width direction of the heat-resistant resin sheet **102** along a longitudinal direction of the heat-resistant resin sheet **102**, turns around and serpentine back in the opposite direction in the other half of the heat-resistant resin sheet **102** in the width direction of the heat-resistant resin sheet **102** along the longitudinal direction of the heat-resistant resin sheet **102** for the above-described length. The external terminals are crimped to both ends of the heat generating wire and power is supplied to the heat generating wire via the external terminals.

FIG. 4B illustrates an example wiring of the heat generator **103** having the linear shape in which heat is generated in a center portion and both end portions of the heat generator **103** in the longitudinal direction of the heat-resistant resin sheet **102**. The both end portions of the heat generator **103** oppose each other via the center portion. Heat may be generated in the center portion or in the center portion and the both end portions of the heat generator **103**.

As illustrated in FIG. 4B, the heat generator **103** includes heat generating wires **103a** and/or **103b** for independently carrying an electric current. The heat generating wire **103a** serpentine by using an almost full width in the width direction of the heat-resistant resin sheet **102** in the center portion

11

of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**. The heat generating wire **103a** extends straight in one half of the heat-resistant resin sheet **102** in the width direction of the heat-resistant resin sheet **102** along the longitudinal direction of the heat-resistant resin sheet **102** in both end portions of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**.

The heat generating wire **103b** serpentine by using the almost full width in the width direction of the heat-resistant resin sheet **102** in the both end portions of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**. The heat generating wire **103b** extends straight in the other half of the heat-resistant resin sheet **102** in the width direction of the heat-resistant resin sheet **102** along the longitudinal direction of the heat-resistant resin sheet **102** in the center portion of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**. The external terminals are crimped to both ends of each of the heat generating wires **103a** and **103b** and power is independently supplied to each of the heat generating wires **103a** and **103b** via the external terminals. Therefore, the heat generating wires **103a** and **103b** are neither overlapped nor crossed on the heat-resistant resin sheet **102**. The heat generating wire **103a** generates heat in the center portion of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**. The heat generating wire **103b** generates heat in the both end portions of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**.

According to the example wiring of the heat generator **103** illustrated in FIG. 4B, the heat generating area of the external heater **101** may be switched between two heated areas in accordance with the size of a sheet S conveyed into the fixing unit **11**. Specifically, when the fixing unit **11** fixes a toner image on a small size sheet S, e.g., only the heat generating wire **103a** is selected to generate heat to heat the center portion of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**. Thus, an electric current may be supplied to the heat generating wire **103a** only, resulting in reduced power consumption. When the fixing unit **11** fixes a toner image on a large size sheet S, an electric current is supplied to both of the heat generating wires **103a** and **103b** so that the heat generating wires **103a** and **103b** may generate heat to simultaneously heat the center portion and the both end portions of the heat-resistant resin sheet **102** in the longitudinal direction of the heat-resistant resin sheet **102**.

The heat generating area of the external heater **101** may be divided into more than two areas to cope with various sheet sizes. Namely, the heat generating area of the external heater **101** may be divided into multiple areas in accordance with the size of sheets S which further vary depending on the orientation of the sheets S. When the heat generating area of the external heater **101** is divided into multiple areas, the heat generating area may be changed in accordance with the size of the sheet S, resulting in further reduced power consumption. The heat generating area may be more precisely adjusted to match with the size of the sheet S. An electric current may not be supplied to the heat generating wire provided in the area which is not used for fixing, resulting in further reduced power consumption.

As illustrated in FIG. 3, the heat insulator **120** contacts and covers an outer circumferential surface of the external heater **101**, which does not face the heating roller **31**. The heat insulator **120** covers the entire outer circumferential surface of the external heater **101**. Namely, the outer circumferential

12

surface of the external heater **101** is not exposed. Thus, heat generated by the external heater **101** may be effectively transferred to the heating roller **31**. Specifically, the heat insulator **120** blocks heat generated by the external heater **101**, transferring the heat in a direction to move away from the heating roller **31**. Exposure of other components to heat radiation from the outer circumferential surface of the external heater **101** accordingly may be reduced if not prevented. The heat generated by the external heater **101** may be mostly directed to the heating roller **31** and may effectively heat the heating roller **31**.

As illustrated in FIG. 5, the heat insulator **120** includes a core **121** and/or a cover **122**. The cover **122** includes a laminated film. Covering the core **121** with the cover **122** in vacuum produces the heat insulator **120**. The core **121** includes a porous material (e.g., urethane foam) and/or a powder material (e.g., silica). The core **121** is sealed under an environment having a vacuum in a range of from about 1 Pa to about 200 Pa. The core **121** includes a material having a strength enduring an atmospheric pressure of an environment. The core **121** does not have a solid structure but has a porous structure including a plenty of cells and includes the porous material (e.g., urethane foam) which may be flexibly deformed.

Otherwise, the core **121** has a porous structure including gaps between powder particles. The core **121** is formed in a plate-like shape having dimensions substantially common to outside dimensions of the external heater **101**. The core **121** may be flexibly deformed in a thickness direction. The cover **122** may endure a difference between the vacuum and the atmospheric pressure and is formed in a thin film-like shape having a gas barrier property capable of blocking transmission of outside air. Covering and sealing the core **121** with the cover **122** in vacuum produces the heat insulator **120**. Inner space of the heat insulator **120** not occupied by the core **121** is vacuum. Namely, the entire inner space of the heat insulator **120** is uniformly vacuum.

Heat insulation effectiveness of the heat insulator **120** depends on heat insulation of the vacuum space rather than on the heat insulation property of the materials. Therefore, the heat insulator **120** may have a thinner, lighter structure while having a sufficient heat insulation property. The thinner heat insulator **120** may be disposed in the limited space in the fixing unit **11** by occupying a reduced space. Thus, the fixing unit **11** need not have a large size, saving space in the image forming apparatus **1**.

The heat insulator **120** having the vacuum structure contacts the outer circumferential surface of the external heater **101** which does not face the heating roller **31**. Thus, the heat insulator **120** may block heat transferred from the outer circumferential surface of the external heater **101** on a shortest route to an interior wall of the movable housing **130**. Heat radiated from an inner circumferential surface of the external heater **101** which faces the heating roller **31** is transferred to the heating roller **31** and is blocked by the heating roller **31**. Thus, the heat insulator **120** may effectively insulate heat.

The heat insulator **120** has an increased flexibility because the cover **122** formed in the thin film-like shape covers the core **121** including the urethane foam which easily bends and/or the fluid powder material. The heat insulator **120** may be deformed in accordance with the outer circumferential surface of the external heater **101** while having the vacuum structure. The heat insulator **120** may uniformly contact the outer circumferential surface of the external heater **101** without an air space formed between the heat insulator **120** and the external heater **101**, providing high heat insulation effectiveness. The heat insulator **120** may uniformly provide the high

13

heat insulation effectiveness on the entire outer circumferential surface of the external heater 101. Therefore, high heat radiation effectiveness may also be provided uniformly between the external heater 101 and the heating roller 31.

As illustrated in FIG. 6A, the fixing unit 11 further includes a spring 131 and/or rollers 132. The spring 131 applies a force moving the movable housing 130 holding the external heater 101 toward the heating roller 31. A motor (not shown) drives and rotates the rollers 132 to move the external heater 101 away from the heating roller 31 against the force applied by the spring 131.

Specifically, a guide (not shown) supports or holds the movable housing 130 holding the external heater 101 in a manner that the movable housing 130 may move in directions E. The compressed spring 131 is disposed to contact the movable housing 130 in a state that a free end of the spring 131 contacts a back outer surface of the movable housing 130 which does not face the heating roller 31 but faces an opposite direction of the heating roller 31. A resilient restoration force of the spring 131 presses the back outer surface of the movable housing 130 to move the movable housing 130 holding the external heater 101 toward the heating roller 31.

The rollers 132 are disposed to contact the movable housing 130 in a state that outer circumferential surfaces of the rollers 132 contact a front outer surface of the movable housing 130 disposed in an opposite side of the back outer surface. The rollers 132 may rotate for an arbitrary angle. Each of the rollers 132 is formed in an oval-like shape and includes a core disposed to deviate from a center of the oval and a shaft fixed to the core. A motor (not shown) rotates the shaft in both clockwise and counterclockwise directions for an arbitrary angle. The rollers 132 rotate for an arbitrary angle, and then stop. A distance for which the rollers 132 protrude toward the movable housing 130 may vary depending on the arbitrary angle. Positions at which the outer circumferential surfaces of the rollers 132 contact the front outer surface of the movable housing 130 move in the directions E within a length determined based on a difference between longest and shortest diameters of the rollers 132. As a result, the external heater 101 held by the movable housing 130 may arbitrarily contact and separate from the heating roller 31.

An actuator for driving the rollers 132 in a rotating direction and/or a driver for driving the rollers 132 in protruding and receding directions may move the movable housing 130 holding the external heater 101 to adjust the distance for which the rollers 132 protrude toward the movable housing 130. The actuator may include a motor and the driver may include a solenoid for contacting the movable housing 130 to move the movable housing 130 by pushing and pulling the movable housing 130.

As described above, a distance between the movable housing 130 and the heating roller 31 may be changed by setting an angle for which the rollers 132 rotate. Namely, the external heater 101 held by the movable housing 130 may move for a length corresponding to the above-described distance with respect to the heating roller 31 for rotating in a state that an axis of the heating roller 31 is fixed. Thus, a distance between the external heater 101 and the heating roller 31 may be arbitrarily changed. The distance between the external heater 101 and the heating roller 31 may be zero, that is, the external heater 101 may contact the heating roller 31.

To fix a toner image on a sheet S, the heating roller 31 rotates and applies heat to the sheet S having the toner image. Therefore, the external heater 101 recedes to a position where the external heater 101 does not contact the rotating heating roller 31 as illustrated in FIG. 6A and thereby may not damage the heating roller 31. Radiant heat is transferred from the

14

external heater 101 to the heating roller 31 to heat the heating roller 31 cooled after heat is transferred from the heating roller 31 to the sheet S. Thus, while the toner image on the sheet S is fixed, the external heater 101 supplies heat to the heating roller 31 in an amount at least sufficient to retrieve heat transferred from the heating roller 31 to the sheet S for fixing the toner image on the sheet S.

After the fixing operation is finished, the heating roller 31 stops rotating in a standby mode for waiting for a next fixing operation. The heating roller 31 is used to maintain a certain temperature in the standby mode. The external heater 101 contacts the heating roller 31 as illustrated in FIG. 6B after the heating roller 31 stops rotating so that heat is directly transferred from the external heater 101 to the heating roller 31. The external heater 101 moves to contact the heating roller 31 to obtain a high heat transfer efficiency.

As described above, according to this example embodiment, heat may be effectively transferred from the external heater 101 to the heating roller 31 in a state that the external heater 101 does not contact the heating roller 31. As a result, warm-up of the heating roller 31 may be finished in a shortened time period. The heating roller 31 may not be cooled down immediately after the warm-up is finished. The external heater 101 may not damage the outer circumferential surface of the heating roller 31.

Specifically, while the heating roller 31 stops rotating before the warm-up is finished, the external heater 101 contacts the heating roller 31, providing the high heat transfer efficiency and shortening the warm-up time period. When the toner image on the sheet S is fixed immediately after the warm-up is finished, that is, when the sheet S having the toner image is conveyed through the nip formed between the heating roller 31 and the pressure roller 32, heat is transferred from the external heater 101 which is disposed along the outer circumferential surface of the heating roller 31 with a distance provided between the external heater 101 and the heating roller 31 so that the external heater 101 does not contact the heating roller 31. Heat may be effectively transferred from the external heater 101 to the heating roller 31 when heat is transferred from the heating roller 31 to the sheet S for fixing. Thus, the heating roller 31 may not be cooled down when the sheet S is conveyed through the nip formed between the heating roller 31 and the pressure roller 32 immediately after the warm-up of the heating roller 31 is finished. Heat is transferred from the external heater 101 to the heating roller 31 in a state that the external heater 101 does not contact the heating roller 31. The outer circumferential surface of the heating roller 31 may not wear due to the contact of the external heater 101 to the heating roller 31, resulting in an improved endurance of the heating roller 31. The external heater 101, from which heat may be effectively transferred, may have a compact size and save space in the fixing unit 11.

The fixing unit 11 includes the movable housing 130 for holding the external heater 101, resulting in easier assembly and maintenance of the fixing unit 11. Specifically, the external heater 101 may be installed into the fixing unit 11 after the external heater 101 is attached to the movable housing 130. The external heater 101 may be deformed. The resin member 140 is attached to the movable housing 130 and the external heater 101 is attached to the movable housing 130 via the resin member 140. Even when the external heater 101 is not molded into the above-described shape, the resin member 140 deforms the external heater 101 when the external heater 101 is attached to the movable housing 130. Thus, a manufacturing process for deforming the external heater 101 may not be needed. When the external heater 101 has a failure, the movable housing 130 holding the external heater 101 may be

15

removed from the fixing unit 11 and the movable housing 130 holding a new external heater 101 may be installed into the fixing unit 11. Thus, the elements of the fixing unit 11 may be easily replaced, resulting in an improved maintenance of the fixing unit 11.

According to this example embodiment, the external heater 101 contacts the heating roller 31 when the heating roller 31 stops and separates from the heating roller 31 when the heating roller 31 moves. Namely, when the heating roller 31 stops, the external heater 101 contacts the heating roller 31 to effectively transfer heat from the external heater 101 to the heating roller 31 contacted thereto. When the heating roller 31 moves, the external heater 101 separates from the heating roller 31 to prevent the external heater 101 from damaging the outer circumferential surface of the heating roller 31. Thus, heat may be effectively transferred from the external heater 101 to the heating roller 31. As a result, the warm-up time of the heating roller 31 may be shortened and the heating roller 31 may not be cooled down immediately after the warm-up of the heating roller 31 is finished. Further, the external heater 101 may not damage the outer circumferential surface of the heating roller 31.

FIGS. 7A and 7B illustrate a fixing unit 11a according to another example embodiment of the present invention. As illustrated in FIG. 7A, the fixing unit 11a includes a heating roller 31a and an external heater 101a respectively replacing the heating roller 31 and the external heater 101 of the fixing unit 11. The other elements of the fixing unit 11a are common to the fixing unit 11. The fixing unit 11a is configured to provide higher heat transfer efficiency than the fixing unit 11 when the external heater 11a contacts the heating roller 31a.

The heating roller 31a functions as a fixing member for applying heat to a sheet S having a toner image. The external heater 101a may contact and separate from the heating roller 31a. When the heating roller 31a stops rotating and the external heater 101a contacts the heating roller 31a, heat is effectively and rapidly transferred from the external heater 101a to the heating roller 31a. However, when an air space is provided between the external heater 101a and the heating roller 31a, heat transfer efficiency may substantially decrease even if the air space is small. In addition, the temperature of an outer circumferential surface of the heating roller 31a may vary depending on whether the air space is provided between the external heater 101a and the heating roller 31a or not. Namely, the temperature of the outer circumferential surface of the heating roller 31a which contacts the external heater 101a may differ from the temperature of the outer circumferential surface of the heating roller 31a which does not contact the external heater 101a, causing temperature variations. As a result, the outer circumferential surface of the heating roller 31a may be easily damaged.

To solve the above-described problems, an inner circumferential surface of the external heater 101a, which contacts the outer circumferential surface of the heating roller 31a, is formed in a concave-like shape (e.g., an arc-like shape when the heating roller 31a is formed in a roller-like shape) which fits to a convex-like shape of the outer circumferential surface of the heating roller 31a. Thus, the inner circumferential surface of the external heater 101a uniformly contacts the outer circumferential surface of the heating roller 31a. Specifically, a radius R101 forming the concave-like shape of the external heater 101a has a length common to a radius R31 forming the convex-like shape of the heating roller 31a. When the external heater 101a contacts the heating roller 31a, the inner circumferential surface of the external heater 101a entirely contacts the outer circumferential surface of the heating roller 31a. The inner circumferential surface of the exter-

16

nal heater 101a which faces the heating roller 31a is formed in a shape corresponding to the shape of the outer circumferential surface of the heating roller 31a. The shapes of the inner circumferential surface of the external heater 101a and the outer circumferential surface of the heating roller 31a cause the external heater 101a to contact the heating roller 31a with no space provided between the external heater 101a and the heating roller 31a.

As illustrated in FIG. 7B, when the external heater 101a contacts the heating roller 31a, the entire inner circumferential surface of the external heater 101a may uniformly contact the outer circumferential surface of the heating roller 31a with no small air space partially created between the external heater 101a and the heating roller 31a. Thus, heat transfer efficiency may not decrease due to the air space, resulting in improved heating efficiency of the external heater 101a. The outer circumferential surface of the heating roller 31a may be uniformly heated and the temperature of the outer circumferential surface of the heating roller 31a may not vary, resulting in improved endurance of the heating roller 31a.

As described above, according to this example embodiment, the fixing unit 11a may provide effects similar to the effects provided by the fixing unit 11 according to a previous example embodiment. Further, when the heating roller 31a stops rotating, the external heater 101a contacts the heating roller 31a by using its entire inner circumferential surface with no air space provided between the external heater 101a and the heating roller 31a. The heating roller 31a may be heated by utilizing the entire contacted surfaces of the external heater 101a and the heating roller 31a. The temperature of the external heater 101a and the heating roller 31a may not vary on the contacted surfaces. Thus, heat may be effectively transferred from the external heater 101a to the heating roller 31a.

According to this example embodiment, the heating roller 31a is formed in the convex-like shape with respect to the inner circumferential surface of the external heater 101a and the external heater 101a is formed in the concave-like shape with respect to the outer circumferential surface of the heating roller 31a. The external heater 101a and the heating roller 31a contact each other in a state that the contacted surfaces of the external heater 101a and the heating roller 31a form the common arc-like shape. When the heating roller 31a stops, the external heater 101a and the heating roller 31a may contact each other to form the contacted surfaces as large as possible. Thus, heat may be effectively and uniformly transferred from the external heater 101a to the heating roller 31a without causing varied temperatures of the outer circumferential surface of the heating roller 31a.

FIG. 8 illustrates a timing chart for controlling the heating roller 31a and the external heater 101a according to yet another example embodiment of the present invention. According to this example embodiment, whether the heating roller 31a is rotated or not is detected. The external heater 101a contacts the heating roller 31a after the heating roller 31a stops rotating and the external heater 101a separates from the heating roller 31a before the heating roller 31a starts rotating, based on the detection.

Specifically, a motor (not shown) for driving and rotating the heating roller 31a rotates and stops in accordance with a rotating signal. A motor (not shown) for contacting and separating the external heater 101a to and from the heating roller 31a is controlled to operate in association with the rotating signals.

As illustrated in FIG. 8, a signal voltage of the rotating signal for controlling the heating roller 31a is switched between low and high states to rotate and stop the motor for

17

driving and rotating the heating roller **31a**. The signal voltage in the low state indicates an ON signal for starting and continuing rotating the motor. The signal voltage in the high state indicates an OFF signal for stopping rotating the motor.

The external heater **101a** contacts and separates from the heating roller **31a** in accordance with a contacting-separating signal. A signal voltage of the contacting-separating signal is switched between low and high states to rotate and stop the motor for contacting and separating the external heater **101a** to and from the heating roller **31a**. The signal voltage in the low state indicates an On signal for rotating the motor forward. The signal voltage in the high state indicates an OFF signal for rotating the motor backward for a length which may be determinable and/or determined.

A timing to start separating the external heater **101a** from the heating roller **31a**, which is indicated with an edge signal of the contacting-separating signal switching from the high state to the low state, is set to a timing which is for a time period F earlier than a timing to start rotating the heating roller **31a**, which is indicated with an edge signal of the rotating signal switching from the low state to the high state. On the other hand, a timing to start contacting the external heater **101a** to the heating roller **31a**, which is indicated with an edge signal of the contacting-separating signal switching from the low state to the high state, is set to a timing which is delayed for a time period G from a timing to stop rotating the heating roller **31a**, which is indicated with an edge signal of the rotating signal switching from the high state to the low state.

The timing to start separating the external heater **101a** from the heating roller **31a** may be controlled by setting the timing to start rotating the heating roller **31a** and outputting the edge signal for starting separating the external heater **101a** from the heating roller **31a** at the timing which is for the time period F prior to the timing to start rotating the heating roller **31a**. Otherwise, the timing to start separating the external heater **101a** from the heating roller **31a** may be controlled by setting the timing to start separating the external heater **101a** from the heating roller **31a** and outputting the edge signal for starting rotating the heating roller **31a** at the timing which is for the time period F delayed from the timing to start separating the external heater **101a** from the heating roller **31a**. The timing to start contacting the external heater **101a** to the heating roller **31a** may be controlled by outputting the edge signal for starting contacting the external heater **101a** to the heating roller **31a** at the timing which is delayed for the time period G from the timing to stop rotating the heating roller **31a**.

The time period F is configured to be equivalent to or longer than a time period needed for the external heater **101a**, which contacts the heating roller **31a**, to separate from the heating roller **31a**. The time period G is configured to be equivalent to or longer than a time period needed for the rotating heating roller **31a** to completely stop rotating.

The external heater **101a** contacts the heating roller **31a** after the heating roller **31a** stops rotating and separates from the heating roller **31a** before the heating roller **31a** starts rotating. Therefore, damage caused by friction between the moving external heater **101a** and the rotating heating roller **31a** may be reduced if not prevented. The external heater **101a** contacts and separates from the heating roller **31a** while the heating roller **31a** completely stops. Thus, the external heater **101a** and the heating roller **31a** may not move in a relatively oblique direction to each other, reducing if not preventing abrasion on the outer circumferential surface of the heating roller **31a** and maintaining the smooth outer circumferential surface of the heating roller **31a**. The smooth

18

outer circumferential surface of the heating roller **31a** may provide proper fixing of a toner image on a sheet S, resulting in forming a high quality image on the sheet S. The damage on the outer circumferential surface of the heating roller **31a** may be reduced if not prevented, resulting in an improved endurance of the heating roller **31a**.

To control the external heater **101a** and the heating roller **31a** to operate as described above, a control circuit (not shown) may be provided in the fixing unit **11a**. Otherwise, a controller (not shown) of the image forming apparatus **1** may be utilized. The control circuit and the controller may be connected to various sensors and switches used for controlling the external heater **101a** and the heating roller **31a** to obtain signals from the sensors and switches. The control circuit and the controller may also be connected to the external heater **101a** and the heating roller **31a** to send commands to the external heater **101** and the heating roller **31a**.

As described above, this example embodiment may provide effects similar to the effects provided according to the previous example embodiment. Further, the outer circumferential surface of the heating roller **31a** may not be damaged even if the external heater **101a** contacts and separates from the heating roller **31a**. The external heater **101a** neither contacts nor separates from the heating roller **31a** while the heating roller **31a** rotates. Namely, the external heater **101a** and the heating roller **31a** do not move simultaneously. Sufficient time periods are provided before the external heater **101a** contacts the heating roller **31a** after the heating roller **31a** stops rotating and before the heating roller **31a** starts rotating after the external heater **101a** separates from the heating roller **31a**, allowing a mechanical error and preventing simultaneous movement of the external heater **101a** and the heating roller **31a**. Thus, abrasion, which may occur on the outer circumferential surface of the heating roller **31a** when the external heater **101a** and the heating roller **31a** contact each other while simultaneously moving, may be reduced if not prevented. As a result, the outer circumferential surface of the heating roller **31a** may be maintained in a proper condition, providing high fixing performance.

According to this example embodiment, the external heater **101a** contacts the heating roller **31a** after the heating roller **31a** stops rotating and separates from the heating roller **31a** before the heating roller **31a** starts rotating, reducing if not preventing abrasion on the outer circumferential surface of the heating roller **31a** caused by friction between the external heater **101a** and the rotating heating roller **31a** contacting each other.

The following describes yet another example embodiment of the present invention. According to this example embodiment, the external heater **101a** contacts the heating roller **31a** when the image forming apparatus **1** is powered off. Specifically, a force applied (e.g., the spring **131**) pushes the external heater **101a** toward the heating roller **31a**. A driver (e.g., a motor for driving the external heater **101a**) applies a force to separate the external heater **101a** from the heating roller **31a**. When the image forming apparatus **1** is powered off, the force for separating the external heater **101a** from the heating roller **31a** is released to contact the external heater **101a** to the heating roller **31a**.

When the image forming apparatus **1** is powered on after powered off (e.g., when the image forming apparatus **1** is warmed up), the external heater **101a** operates after power is supplied to all devices provided in the image forming apparatus **1**. Before the external heater **101a** contacts the heating roller **31a**, a time period for sending a signal to the driver of the external heater **101a** and a time period for moving the

19

external heater 101a to the heating roller 31a are needed, lengthening a warm-up time of the heating roller 31a.

If the external heater 101a is configured to contact the heating roller 31a in an initial state (e.g., when the image forming apparatus 1 is powered on), the heating roller 31a may be heated in a short time period, shortening the warm-up time of the heating roller 31a. Specifically, when the image forming apparatus 1 is powered off, the driver for driving the external heater 101a is powered off and the external heater 101a contacts the heating roller 31a. Therefore, when the image forming apparatus 1 is powered on again (e.g., when the warm-up starts), the external heater 101a contacts the heating roller 31a.

Thus, the time period for sending a signal to the driver of the external heater 101a and the time period for moving the external heater 101a to the heating roller 31a may be reduced or eliminated, resulting in a shorter warm-up time of the heating roller 31a. In other words, when the image forming apparatus 1 is powered off, the driver of the external heater 101a does not drive the external heater 101a or reduces its driving force resisting the spring 131 to a level which causes the external heater 101a to contact the heating roller 31a. A force (e.g., an elastic force) of the spring 131 moves the external heater 101a (i.e., the movable housing 130 holding the external heater 101a) toward the heating roller 31a so that the external heater 101a contacts the heating roller 31a.

As described above, this example embodiment may provide effects similar to the effects provided according to the previous example embodiment. Further, when the image forming apparatus 1 is powered on again after powered off, power supply to the external heater 101a immediately starts to heat the external heater 101a. Heat may be directly transferred from the external heater 101a to the heating roller 31a, shortening a first warm-up time for heating the heating roller 31a from a room temperature to a fixing temperature.

When the image forming apparatus 1 is powered on again after the image forming apparatus 1 is powered off and the temperature of the heating roller 31a decreases to the room temperature, the heating roller 31a may be heated in a short time period, shortening the warm-up time of the heating roller 31a. Even when the image forming apparatus 1 is powered on early in the morning when the room temperature is substantially low, for example, the heating roller 31a may be heated in a short time period. As a result the image forming apparatus 1 waits for the heating roller 31a to be warmed up for the short time period, providing an improved user convenience.

According to this example embodiment, when the image forming apparatus 1 is powered off, the external heater 101a moves to contact the heating roller 31a. Namely, the external heater 101a is controlled to contact the heating roller 31a in the initial state in which the image forming apparatus 1 is powered off. Thus, when the image forming apparatus 1 is powered on again, power is immediately supplied to the external heater 101a so that the external heater 101a generates heat. The generated heat may be directly transferred from the external heater 101a to the heating roller 31a contacted thereto, shortening the warm-up time of the heating roller 31a.

The following describes yet another example embodiment of the present invention. As described above, in the fixing unit 11 according to the preceding example embodiment, the external heater 101 contacts the heating roller 31 only while the heating roller 31 stops rotating to effectively transfer heat from the external heater 101 to the heating roller 31. In the standby mode when a sheet S is not conveyed in the fixing unit 11 and heat capacity of the heating roller 31 and the fixing unit 11 increases closer to a saturation level, less power, which is

20

sufficient to maintain the temperature of the heating roller 31 in the standby mode, is supplied to the external heater 101.

When power is not supplied to the external heater 101 in the standby mode, the external heater 101 draws heat from the heating roller 31. To prevent this, e.g., the external heater 101 is configured to contact the heating roller 31 only when power is supplied to the external heater 101. When the external heater 101 contacts the heating roller 31, heat is effectively transferred from the external heater 101 to the heating roller 31. However, when the external heater 101 contacts the heating roller 31 while power is not supplied to the external heater 101 in the standby mode, the external heater 101 may cause the heating roller 31 to radiate heat. To prevent this, e.g., the external heater 101 is configured not to contact the heating roller 31 when power is not supplied to the external heater 101.

When the image forming apparatus 1 or at least the fixing unit 11 is in the standby mode, that is, when power is not supplied to the external heater 101, the external heater 101 separates from the heating roller 31. The contacting-separating signal is output to contact the external heater 101 to the heating roller 31, e.g., only when power is supplied to the external heater 101. When power is supplied to the external heater 101, the contacting-separating signal is continuously output to contact the external heater 101 to the heating roller 31 so that heat is directly transferred from the external heater 101 to the heating roller 31, contacted thereto. When power is not supplied to the external heater 101, output of the contacting-separating signal immediately stops and the external heater 101 moves in a direction in which the external heater 101 separates from the heating roller 31.

In the fixing unit 11 configured as described above, the external heater 101 separates from the heating roller 31 e.g., when power is not supplied to the external heater 101 in the standby mode and contacts the heating roller 31 only when power is supplied to the external heater 101. Thus, when the external heater 101 does not heat the heating roller 31 while power is not supplied to the external heater 101, heat may not be easily transferred from the heating roller 31 to the external heater 101. Namely, heat transfer from the heating roller 31 to the external heater 101 having a greater heat capacity may be suppressed.

When power is not supplied to the external heater 101, that is, when the external heater 101 does not generate heat after the external heater 101, to which power is supplied, heats the heating roller 31, heat may not be radiated from the heating roller 31 to the external heater 101. Thus, heat may be effectively utilized. As a result, power consumed by the external heater 101 to generate heat may be reduced. Further, endurance of the external heater 101 may improve. Specifically, it may take longer for the external heater 101 to be cooled down to a desired temperature. Namely, it may take a long time for the temperature of the external heater 101 to change, resulting in a reduced frequency for supplying power to the external heater 101 to cause the external heater 101 to generate heat.

As described above, this example embodiment may provide effects similar to the effects provided according to the previous example embodiment. Further, when power is not supplied to the external heater 101 and thereby the external heater 101 does not generate heat, the external heater 101 separates from the heating roller 31 to suppress heat transfer from the heating roller 31 to the external heater 101. Namely, heat stored on the heating roller 31 may not easily be transferred to the external heater 101 having the greater heat capacity.

According to this example embodiment, the external heater 101 moves to separate from the heating roller 31 when power

21

is not supplied to the external heater **101**. Namely, the external heater **101** is controlled to separate from the heating roller **31** regardless of movement of the heating roller **31**, when power is not supplied to the external heater **101**. Thus, heat transfer from the heating roller **31** to the external heater **101** may be suppressed while the external heater **101** does not heat the heating roller **31**. Namely, heat stored on the heating roller **31** may be transferred to the external heater **101** having the greater heat capacity as little as possible.

FIGS. **9** and **10** illustrate yet another example embodiment of the present invention. As illustrated in FIG. **9**, an image forming apparatus **1b** includes a fixing unit **11b** instead of the fixing unit **11** illustrated in FIG. **1**. The fixing unit **11b** includes a belt-like shape fixing member for applying heat to a sheet **S** having a toner image. The other elements of the image forming apparatus **1b** are common to the image forming apparatus **1** illustrated in FIG. **1**.

As illustrated in FIG. **10**, the fixing unit **11b** includes the external heater **101**, the heat insulator **120**, the resin member **140**, a first roller **201**, a second roller **202**, a fixing belt **203**, a pressure roller **204**, and/or a unit housing **230**.

The fixing belt **203** is looped over the first roller **201** and the second roller **202**. The pressure roller **204** opposes the second roller **202** via the fixing belt **203** and functions as a pressing member for applying pressure to the second roller **202** via the fixing belt **203**. The external heater **101** is disposed along an outer circumferential surface of the first roller **201** via the fixing belt **203** with a distance provided between the external heater **101** and the fixing belt **203** opposing each other. The external heater **101** is disposed to form a concave-like shape with respect to the outer circumferential surface of the first roller **201** opposing thereto via the fixing belt **203** and does not contact the fixing belt **203**. The heat insulator **120** is disposed to contact the outer circumferential surface of the external heater **101**, which does not face the fixing belt **203**.

In the fixing unit **11b**, bearings (not shown) rotatably support a shaft of the second roller **202** extended in the horizontal direction. The first roller **201** is disposed away from the second roller **202** with a distance provided between the first roller **201** and the second roller **202** in a substantially horizontal direction. Bearings (not shown) rotatably support a shaft of the first roller **201** extended in the horizontal direction. The bearings of the first roller **201** and the second roller **202** are disposed parallel to each other. The first roller **201** is formed in a hollow cylinder-like shape having a diameter shorter than a diameter of the second roller **202**. The fixing belt **203** is looped over the first roller **201** and the second roller **202**. The fixing belt **203** is formed in an endless belt shape and functions as a fixing member for applying heat to a sheet **S** having a toner image. A driver including a motor (not shown) drives and rotates the second roller **202** in a rotating direction **H** at a constant speed. The rotating second roller **202** drives and rotates the fixing belt **203** in a rotating direction **I** at a constant speed. The rotating fixing belt **203** drives and rotates the first roller **201** in a rotating direction **J**.

The pressure roller **204** opposes the second roller **202** via the fixing belt **203**. Bearings (not shown) rotatably support a shaft of the pressure roller **204** extended in the horizontal direction. The shaft of the pressure roller **204** may move toward the shaft of the second roller **202** for a distance. A presser (not shown) including the pressing lever **26** illustrated in FIG. **2** presses the pressure roller **204** toward the second roller **202**. An outer circumferential surface of the pressure roller **204** contacts a portion on an outer circumferential surface of the fixing belt **203**, which is looped over the second roller **202**. The outer circumferential surfaces of the pressure roller **204** and the fixing belt **203** contacting each other form

22

a nip (e.g., a fixing position) to which a pressure is applied. A toner image on a sheet **S** is fixed while the sheet **S** is conveyed through the nip.

The fixing belt **203** has a width corresponding to a maximum sheet size the image forming apparatus **1b** may handle. The fixing belt **203** has flexibility and tensile strength needed for a belt as well as heat resistance, thermal conductivity, and compression strength in a thickness direction needed for fixing using heat and pressure. The fixing belt **203** includes a base, an elastic layer, and/or a releasing layer. The base includes a heat resistant resin and has a thickness in a range of from about 30 μm to about 100 μm based on a balance between thermal conductivity and strength. The elastic layer is disposed under the releasing layer and includes a heat resistant rubber (e.g., a silicone rubber and a fluorocarbon rubber) which causes the outer circumferential surface of the fixing belt **203** to uniformly contact a toner image formed on a sheet **S**. The releasing layer is disposed to cover the elastic layer and includes fluoroplastic which provides releasing and heat resistance properties because the fixing belt **203** pressingly contacts a sheet **S** and a toner image formed on the sheet **S**. Thus, the fixing belt **203** rotates to receive heat and stably carry heat to the fixing position.

The external heater **101** is deformed to fit along an outer circumferential surface of the first roller **201**. The heat insulator **120** contacts and entirely covers the outer circumferential surface of the external heater **101** which does not face the first roller **201** via the fixing belt **203**. The unit housing **230** forms a part of the housing **70** illustrated in FIG. **2**. The resin member **140** is attached to the unit housing **230**. The external heater **101** and the heat insulator **120** are directly attached to the resin member **140**. Thus, the external heater **101** is disposed between the housing **70** and the first roller **201**.

As described above, according to this example embodiment, the external heater **101** is disposed along the outer circumferential surface of the first roller **201** via the fixing belt **203** and includes a concave-like shape portion with respect to the outer circumferential surface of the first roller **201** opposing the external heater **101**. The external heater **101** does not contact the fixing belt **203** and a distance is uniformly provided between the external heater **101** and the fixing belt **203**. Thus, heat may be effectively transferred from the external heater **101** to the fixing belt **203**, thereby shortening a warm-up time of the fixing belt **203** and preventing the fixing belt **203** from being cooled down immediately after the fixing belt **203** is warmed up. Further, the outer circumferential surface of the fixing belt **203** may not be damaged.

The external heater **101** is attached to the unit housing **230** via the resin member **140** and is disposed between the unit housing **230** and the heating belt **203**, resulting in easy maintenance and space saving in the fixing unit **11b**.

According to this example embodiment, the fixing unit **11b** includes the rotatable first roller **201**, the rotatable second roller **202**, the fixing belt **203** looped over the first roller **201** and the second roller **202**, the pressure roller **204** for applying pressure to the second roller **202** via the fixing belt **203**, and/or the external heater **101** disposed along the outer circumferential surface of the first roller **201** via the fixing belt **203**. The external heater **101** is formed in the concave-like shape with respect to the outer circumferential surface of the first roller **201** so as to heat the fixing belt **203** without contacting the fixing belt **203**. The external heater **101** is disposed close to the fixing belt **203** with a uniform distance provided between the external heater **101** and the outer circumferential surface of the fixing belt **203**. Thus, heat may be effectively transferred from the external heater **101** to the fixing belt **203**. As a result, the fixing belt **203** may be warmed up in a short

23

time period and may not be cooled down immediately after the warm-up is finished. Further, the external heater 101 may not damage the outer circumferential surface of the fixing belt 203.

FIG. 11 illustrates a fixing unit 11c according to yet another example embodiment of the present invention. As illustrated in FIG. 11, the fixing unit 11c includes the first roller 201, the second roller 202, the fixing belt 203, the pressure roller 204, an external heater 101c, and/or a heat insulator 120c.

The fixing belt 203 is looped over the first roller 201 and the second roller 202. The pressure roller 204 opposes the second roller 202 via the fixing belt 203 and function as a pressing member for applying pressure to the second roller 202 via the fixing belt 203. The external heater 101c is disposed along an outer circumferential surface of the second roller 202 via the fixing belt 203 with a distance provided between the external heater 101c and the fixing belt 203 opposing each other in a state that the external heater 101c does not contact the fixing belt 203. The heat insulator 120c is disposed to contact an outer circumferential surface of the external heater 101c, which does not face the second roller 202 via the fixing belt 203.

The external heater 101c is deformed to fit along the outer circumferential surface of the second roller 202. The heat insulator 120c contacts and entirely covers the outer circumferential surface of the external heater 101c, which does not face the second roller 202 via the fixing belt 203. The fixing unit 11c further includes a unit housing (not shown) and/or a resin member (not shown). The unit housing forms a part of the housing 70 illustrated in FIG. 2. The resin member is attached to the unit housing. The external heater 101c and the heat insulator 120c are directly attached to the resin member. Thus, the external heater 101c is disposed between the housing 70 and the second roller 202.

The structure of each of the external heater 101c and the heat insulator 120c is common to the external heater 101 and the heat insulator 120 of the fixing unit 11 according to the preceding example embodiment.

As described above, according to this example embodiment, the external heater 101c is disposed along the outer circumferential surface of the second roller 202 via the fixing belt 203 and includes a concave-like shape portion with respect to the outer circumferential surface of the second roller 202 opposing the external heater 101c. The external heater 101c does not contact the fixing belt 203 and a distance is uniformly provided between the external heater 101c and the fixing belt 203. Thus, heat may be effectively transferred from the external heater 101c to the fixing belt 203, thereby shortening a warm-up time of the fixing belt 203 and preventing the fixing belt 203 from being cooled down immediately after the fixing belt 203 is warmed up. Further, the outer circumferential surface of the fixing belt 203 may not be damaged.

The external heater 101c is attached to the unit housing via the resin member and is disposed between the unit housing and the heating belt 203, resulting in easy maintenance and space saving in the fixing unit 11c.

According to this example embodiment, the fixing unit 11c includes the rotatable first roller 201, the rotatable second roller 202, the fixing belt 203 looped over the first roller 201 and the second roller 202, the pressure roller 204 for applying pressure to the second roller 202 via the fixing belt 203, and/or the external heater 101c disposed along the outer circumferential surface of the second roller 202 via the fixing belt 203. The external heater 101c is formed in the concave-like shape with respect to the outer circumferential surface of the second roller 202 so as to heat the fixing belt 203 without

24

contacting the fixing belt 203. The external heater 101c is disposed close to the fixing belt 203 with a uniform distance provided between the external heater 101c and the outer circumferential surface of the fixing belt 203. Thus, heat may be effectively transferred from the external heater 101c to the fixing belt 203. As a result, the fixing belt 203 may be warmed up in a short time period and may not be cooled down immediately after the warm-up is finished. Further, the external heater 101c may not damage the outer circumferential surface of the fixing belt 203.

FIG. 12 illustrates a fixing unit 11d according to yet another example embodiment of the present invention. As illustrated in FIG. 12, the fixing unit 11d includes the external heater 101, the heat insulator 120, the resin member 140, the unit housing 230, the first roller 201, the second roller 202, the fixing belt 203, the pressure roller 204, the external heater 101c, and/or the heat insulator 120c.

The fixing belt 203 is looped over the first roller 201 and the second roller 202. The pressure roller 204 opposes the second roller 202 via the fixing belt 203 and functions as a pressing member for applying pressure to the second roller 202 via the fixing belt 203. The external heater 101 is disposed along the outer circumferential surface of the first roller 201 via the fixing belt 203 with a distance provided between the external heater 101 and the fixing belt 203 opposing each other in a state that the external heater 101 does not contact the fixing belt 203. The external heater 101 includes a concave-like shape portion with respect to the outer circumferential surface of the first roller 201 opposing the external heater 101. The external heater 101c is disposed along the outer circumferential surface of the second roller 202 via the fixing belt 203 with a distance provided between the external heater 101c and the fixing belt 203 opposing each other in a state that the external heater 101c does not contact the fixing belt 203. The heat insulators 120 and 120c are respectively disposed to contact the outer circumferential surfaces of the external heaters 101 and 101c, which do not face the fixing belt 203.

Namely, the fixing unit 11d includes the external heater 101 and its peripheral elements of the fixing unit 111b illustrated in FIG. 10 and the external heater 101c and its peripheral elements of the fixing unit 11c illustrated in FIG. 11.

As described above, according to this example embodiment, the fixing unit 11d includes two external heaters 101 and 101c. The external heaters 101 and 101c are respectively disposed along the outer circumferential surfaces of the first roller 201 and the second roller 202 via the fixing belt 203 and include a concave-like shape portion with respect to the outer circumferential surfaces of the first roller 201 and the second roller 202 opposing the external heaters 101 and 101c. The external heaters 101 and 101c do not contact the fixing belt 203 and a distance is uniformly provided between each of the external heaters 101 and 101c and the fixing belt 203. Thus, heat may be effectively transferred from the external heaters 101 and 101c to the fixing belt 203, thereby shortening a warm-up time of the fixing belt 203 and preventing the fixing belt 203 from being cooled down immediately after the fixing belt 203 is warmed up. Further, the outer circumferential surface of the fixing belt 203 may not be damaged.

According to this example embodiment, two external heaters 101 and 101c heat one fixing belt 203. Each of the external heaters 101 and 101c, when provided together in the fixing unit 11d, may generate heat in a decreased amount and strength compared to when only one of the external heaters 101 and 101c is provided, reducing a load applied to each of the external heaters 101 and 101c and thereby improving endurance of each of the external heaters 101 and 101c. Further, the external heaters 101 and 101c may stably heat the

25

fixing belt 203 with high precision, resulting in improved control of the fixing temperature. Thus, the fixing unit 11d may provide improved fixing performance.

The external heater 101 is provided between the unit housing 230 and the fixing belt 203 and the external heater 101c is provided between the unit housing (not shown) having a common structure to the unit housing of the fixing unit 11c illustrated in FIG. 11 and the fixing belt 203, resulting in space saving in the fixing unit 11d. The external heaters 101 and 101c are respectively attached to the unit housing 230 and the unit housing (not shown), resulting in easy maintenance and space saving in the fixing unit 11d.

According to this example embodiment, the fixing unit 11d includes the rotatable first roller 201, the rotatable second roller 202, the fixing belt 203 looped over the first roller 201 and the second roller 202, the pressure roller 204 for applying pressure to the second roller 202 via the fixing belt 203, and/or two external heaters 101 and 101c respectively disposed along the outer circumferential surfaces of the first roller 201 and the second roller 202 via the fixing belt 203. The external heaters 101 and 101c are respectively formed in the concave-like shape with respect to the outer circumferential surfaces of the first roller 201 and the second roller 202 so as to heat the fixing belt 203 without contacting the fixing belt 203. Each of the external heaters 101 and 101c is disposed close to the fixing belt 203 with a uniform distance provided between each of the external heaters 101 and 101c and the outer circumferential surface of the fixing belt 203. Thus, heat may be effectively transferred from the external heaters 101 and 101c to the fixing belt 203. As a result, the fixing belt 203 may be warmed up in a short time period and may not be cooled down immediately after the warm-up is finished. Further, the external heaters 101 and 101c may not damage the outer circumferential surface of the fixing belt 203.

FIG. 13 illustrates a fixing unit 11e according to yet another example embodiment of the present invention. As illustrated in FIG. 13, the fixing unit 11e includes the first roller 201, the second roller 202, the fixing belt 203, the pressure roller 204, the cleaner 74, the separating nail 75, an external heater 301, cam shafts 302, tension belts 303, and/or a temperature sensor 205.

The fixing belt 203 is looped over the first roller 201 and the second roller 202 and rotates in the rotating direction I. The pressure roller 204 opposes the second roller 202 via the fixing belt 203 and functions as a pressing member for applying pressure to the second roller 202 via the fixing belt 203. The external heater 301 is disposed along the outer circumferential surface of the fixing belt 203 with a distance provided between the external heater 301 and the fixing belt 203 opposing each other in a state that the external heater 301 does not contact the fixing belt 203. The cam shafts 302 are formed in a cam shape and are disposed to contact both ends in the rotating direction I on an outer circumferential surface of the external heater 301, which does not face the fixing belt 203. The cam shafts 302 are not formed in a uniform shape in cross section. Namely, the cam shafts 302 do not have a uniform shape in cross section in a longitudinal direction of the cam shafts 302 and are not formed in a uniformly-round bar shape. The tension belts 303 pull both edges of the external heater 301 in the rotating direction I.

The external heater 301 faces a flat portion on the outer circumferential surface of the fixing belt 203, which is formed between the first roller 201 and the second roller 202 and extends straight for a length in the rotating direction I. In an initial state, the cam shafts 302 support the external heater 301 in a state that an inner circumferential surface of the external heater 301 faces the entire flat portion on the outer

26

circumferential surface of the fixing belt 203 with a distance uniformly provided between the external heater 301 and the fixing belt 203 opposing each other. The external heater 301 may be flexibly deformed. The cam shafts 302 may be rotated through an angle. Thus, the external heater 301 may be entirely moved closer to the flat portion on the outer circumferential surface of the fixing belt 203. Otherwise, a substantially center portion on the inner circumferential surface of the external heater 301 in a width direction (e.g., a direction perpendicular to the rotating direction I) of the external heater 301, which extends in a longitudinal direction (e.g., the rotating direction I) of the external heater 301 between the cam shafts 302 may be moved closer to the flat portion on the outer circumferential surface of the fixing belt 203. In this case, portions other than the substantially center portion on the inner circumferential surface of the external heater 301 may be positioned away from the flat portion on the outer circumferential surface of the fixing belt 203.

The temperature sensor 205 includes a terminal disposed to contact the outer circumferential surface of the fixing belt 203 to detect a temperature of the outer circumferential surface of the fixing belt 203.

As illustrated in FIGS. 14A and 14B, the external heater 301 includes a heat-resistant resin sheet 102 and/or a heat generator 103. The heat-resistant resin sheet 102 is formed in a sheet-like shape and includes a polyimide film having a thickness in a range of from about 0.5 mm to about 1.0 mm. The heat generator 103 is buried in the heat-resistant resin sheet 102. The heat generator 103 includes an electrode and an external terminal crimped to the electrode. Specifically, the external terminal is connected to the electrode by crimping. FIG. 14A illustrates the heat generator 103 disposed on the entire surface of the heat-resistant resin sheet 102. The external heater 301 illustrated in FIG. 14A has a structure common to the external heater 101 illustrated in FIG. 4A. FIG. 14B illustrates the heat generator 103 linearly disposed in the heat-resistant resin sheet 102. When the temperature of the external heater 301 reaches a reference temperature (which may be based, e.g., upon a resistance suitable for the heat generator 103), the heat generator 103 may be controlled to stop generating heat and thereby the temperature of the external heater 301 does not exceed the reference temperature. Thus, a thermostat, a thermal fuse, and/or a temperature controller may be eliminated.

As illustrated in FIG. 15, the tension belts 303 apply a tension for pressing the external heater 301 toward the cam shafts 302. The external heater 301 may be deformed in accordance with the shape of the cam shafts 302. The cam shafts 302 do not have a uniform shape in cross section in a sub-scanning direction (e.g., the longitudinal direction of the cam shafts 302). Therefore, the entire surface of the external heater 301 or the center portion on the surface of the external heater 301 in the width direction of the external heater 301 may be selectively positioned close to the outer circumferential surface of the fixing belt 203 by rotating the cam shafts 302.

As illustrated in FIGS. 13 and 15, the tension belts 303 continuously apply a tension to the external heater 301 so that outer edges of the external heater 301 are pulled obliquely downward as illustrated in FIG. 13 and stretched outward as illustrated in FIG. 15. Thus, the external heater 301 is deformed in accordance with the outer shape of the cam shafts 302 in the longitudinal direction of the cam shafts 302. Bearings (not shown) support the cam shafts 302 in the fixing unit 11e. A driver (not shown) including a motor drives and rotates the cam shafts 302 through an angle in a rotating direction. The cam shafts 302 may stop at an arbitrary angle. The ends

27

of the tension belts **303**, which may be elastically stretched, are fixed in the fixing unit **11e**. Otherwise, the ends of the tension belts **303** are fixed in the fixing unit **11e** by using elastic members (e.g., springs) which may be elastically stretched.

The two cam shafts **302** are formed in a common shape having a sufficient strength. As illustrated in FIGS. **16A**, **16B**, and **16C**, the cam shaft **302** includes a first ridge **302a** and/or a second ridge **302b**. The first ridge **302a** is formed along the longitudinal direction of the cam shaft **302** and uniformly protrudes for a length in a radial direction of the cam shaft **302**. The second ridge **302b** protrudes for a length in the radial direction of the cam shaft **302** at an angle shifted for about 90 degrees from the first ridge **302a**. The center portion of the second ridge **302b** in the longitudinal direction of the cam shaft **302** protrudes for a length common to the first ridge **302a**.

When at least sheet size is automatically or manually selected, the cam shafts **302** are driven and rotated through an angle in a rotating direction so that either the first ridge **302a** or the second ridge **302b** supports the surface of the external heater **301** in accordance with the size of a sheet **S**. Thus, either the first ridge **302a** or the second ridge **302b** may protrude for a length longer than a length for which any other part on the outer circumferential surface of the cam shaft **302** protrudes toward the flat outer circumferential surface of the fixing belt **203**, which opposes the external heater **301**. The surface of the external heater **301** entirely or partially moves close to the flat outer circumferential surface of the fixing belt **203** in accordance with the sheet size selected. Thus, variations in the temperature of the outer circumferential surface of the fixing belt **203** may be suppressed.

Specifically, when a small size sheet **S** is conveyed through the nip formed between the pressure roller **204** and the fixing belt **203**, the small size sheet **S** contacts and draws heat from a center portion on the outer circumferential surface of the fixing belt **203** in the width direction of the fixing belt **203**. As a result, the temperature of the outer circumferential surface of the fixing belt **203** may vary in the width direction of the fixing belt **203**. To reduce if not prevent this, the cam shafts **302** cause the center portion on the surface of the external heater **301** in the width direction of the external heater **301** to move close to the center portion on the outer circumferential surface of the fixing belt **203**.

To control the external heater **301** to operate as described above, a control circuit (not shown) may be provided in the fixing unit **11e**. Otherwise, a controller (not shown) of the image forming apparatus **1b** may be utilized. The control circuit and the controller may be connected to various sensors and switches used for controlling the external heater **301** to obtain signals from the sensors and switches. The control circuit and the controller may also be connected to the external heater **301** to send commands to the external heater **301**.

In the fixing unit **11e**, the external heater **301** heats a portion on the outer circumferential surface of the fixing belt **203** without contacting the fixing belt **203** immediately before the portion on the outer circumferential surface of the fixing belt **203** reaches the nip formed between the pressure roller **204** and the fixing belt **203**, providing improved heating efficiency. Specifically, the external heater **301** heats the portion on the outer circumferential surface of the fixing belt **203**, which rotates near and upstream of the nip formed between the pressure roller **204** and the fixing belt **203** in the rotating direction **I** without contacting the portion on the outer circumferential surface of the fixing belt **203**. A distance for which the heated portion moves until reaching the nip formed between the pressure roller **204** and the fixing belt **203** may be

28

shortened, preventing or reducing heat transferred from the heated fixing belt **203** to elements other than the second roller **202**. Thus, the improved heating efficiency may provide a small size external heater **301** and reduced power consumption of the fixing unit **11e**.

As described above, according to this example embodiment, the external heater **301** is disposed along the outer circumferential surface of the fixing belt **203** and is formed in a shape for heating the fixing belt **203** along the outer circumferential surface of the fixing belt **203** without contacting the outer circumferential surface of the fixing belt **203**. Thus, the external heater **301** may not damage the outer circumferential surface of the fixing belt **203**. The external heater **301** moves close to the fixing belt **203** to effectively transfer heat to the fixing belt **203**, shortening the warm-up time of the fixing belt **203** and preventing the temperature of the fixing belt **203** from decreasing immediately after the warm-up is finished.

A distance between the entire or partial surface of the external heater **301** and the fixing belt **203** may be arbitrarily changed in accordance with the size of a sheet **S**, suppressing variations in the temperature of the outer circumferential surface of the fixing belt **203** in the width direction of the fixing belt **203** after a small size sheet **S** is conveyed through the nip formed between the pressure roller **204** and the fixing belt **203**. Thus, the outer circumferential surface of the fixing belt **203** may be uniformly heated in the width direction of the fixing belt **203**, improving fixing performance.

A heat insulator may be attached to the external heater **301** to obtain effects provided by the heat insulator **120** of the fixing unit **11**. In this case, for example, the heat insulator may have a structure common to the heat insulator **120** but may have a thickness for not preventing deformation of the external heater **301** in accordance with the size of a sheet **S**. The heat insulator may cover a portion on the outer circumferential surface of the external heater **301**, which does not face the fixing belt **203** and does not contact the cam shafts **302**.

According to this example embodiment, the distance between the external heater **301** and the fixing belt **203** may be changed in the sub-scanning direction. The external heater **301** is disposed along the outer circumferential surface of the fixing belt **203** so as to heat the fixing belt **203** without contacting the fixing belt **203**. The external heater **301** is disposed close to the fixing belt **203** to effectively transfer heat from the external heater **301** to the fixing belt **203** without damaging the outer circumferential surface of the fixing belt **203**. As a result, the fixing belt **203** may be warmed up in a short time period and may not be cooled down immediately after the warm-up of the fixing belt **203** is finished. The distance between the external heater **301** and the fixing belt **203** may be arbitrarily changed in accordance with the size of a sheet **S**. Thus, even after a small size sheet **S** contacts the fixing belt **203** for fixing, the temperature of the fixing belt **203** may not vary in the width direction of the fixing belt **203**.

The cam shafts **302** are disposed on the outer circumferential surface of the external heater **301**, which does not face the fixing belt **203**, to adjust the distance between the external heater **301** and the fixing belt **203**. The cam shafts **302** have the non-uniform shape in cross section in the sub-scanning direction. The cam shafts **302** support the external heater **301** in a state that the tension belts **303** pull the outer edges of the external heater **301**. Thus, the distance between the external heater **301** and the fixing belt **203** may be changed with a simple structure to suppress variations in the temperature of the fixing belt **203** in the width direction of the fixing belt **203**.

According to the example embodiments as described above, the external heaters **101**, **101a**, **101c**, and **301** include the heat-resistant resin sheet **102** and the heat generator **103**

buried in the heat-resistant resin sheet **102**. The external heaters **101**, **101a**, **101c**, and **301** are formed in the sheet-like shape having the thickness not greater than about 1 mm. Thus, the external heaters **101**, **101a**, **101c**, and **301** may save space in the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e**.

The external heaters **101**, **101a**, **101c**, and **301** may be deformed. Therefore, the external heaters **101**, **101a**, **101c**, and **301** may be disposed along the outer circumferential surface of the fixing member so as to effectively transfer heat from the external heaters **101**, **101a**, **101c**, and **301** to the fixing member. The external heaters **101**, **101a**, **101c**, and **301** may be disposed in the limited space to save space in the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e**. Namely, the external heaters **101**, **101a**, **101c**, and **301** may be deformed into an arbitrary shape which fits the portion on the outer circumferential surface of the fixing member, to which the external heaters **101**, **101a**, **101c**, and **301** oppose. The deformed external heaters **101**, **101a**, **101c**, and **301** are disposed along the portion on the outer circumferential surface of the fixing member, to which the external heaters **101**, **101a**, **101c**, and **301** oppose. Thus, the external heaters **101**, **101a**, **101c**, and **301** may be disposed in the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e** by occupying a reduced space.

The external heaters **101**, **101a**, **101c**, and **301** include the heat insulator **120**, including the core and the cover, produced by covering the core with the cover in vacuum. Therefore, the heat insulator **120** may have the thin shape. The thin heat insulator **120** may be disposed in the limited space in the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e** by occupying a reduced space.

The fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e** include the movable housing **130** or the unit housing **230**. The external heaters **101**, **101a**, **101c**, and **301** are disposed between the movable housing **130** or the unit housing **230** and the fixing member, saving space in the fixing units **11**, **11a**, **11b**, **11d**, and **11e**.

The movable housing **130** or the unit housing **230** supports the external heaters **101**, **101a**, **101c**, and **301** and the heat insulator **120**, resulting in easy maintenance and space saving in the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e**.

The image forming apparatuses **1** and **1b** include the fixing unit **11**, **11a**, **11b**, **11c**, **11d**, or **11e**, and thereby provide the above-described effects and improved fixing performance.

According to the above-described example embodiments, the external heater **101** of the fixing unit **11b**, the external heater **101c** of the fixing unit **11c**, and the external heater **101d** of the fixing unit **11d** are not configured to move, but may be configured to move to contact and separate from the fixing belt **203** like the external heater **101** of the fixing unit **11** and to include one or more controls of the fixing units **11** and **11a**. When the external heater **101** of the fixing unit **11b**, the external heater **101c** of the fixing unit **11c**, or the external heater **101d** of the fixing unit **11d** is configured to contact the fixing belt **203**, the surface of the external heater **101**, **101c**, or **101d** which contacts the surface of the fixing belt **203** may be formed in a shape corresponding to the outer circumferential surface of the fixing belt **203** so that the entire surface of the external heater **101**, **101c**, or **101d** may contact the outer circumferential surface of the fixing belt **203**, like the external heater **101a** of the fixing unit **11a**.

According to the above-described example embodiment, the external heater **301** of the fixing unit **11e** is configured to move close to the fixing belt **203** without contacting the fixing belt **203** by using the cam shafts **302**, but may be configured to entirely contact the fixing belt **203** and to include one or more controls of the fixing units **11b**, **11c**, and **11d**.

In the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e**, the heaters (e.g., the external heaters **101**, **101a**, **101c**, and **301**) are disposed outside the fixing members (e.g., the heating rollers **31** and **31a** and the fixing belt **203**). However, an auxiliary heater for heating the fixing member together with the heater may be disposed inside the fixing member. For example, when the heating roller **31** or **31a** is used as the fixing member, the heating roller **31** or **31a** may have a thin thickness and may include a material for effectively transferring heat from an inner circumferential surface to an outer circumferential surface of the heating roller **31** or **31a**. The auxiliary heater (e.g., an electric heater and a halogen lamp) may be disposed inside the heating roller **31** or **31a** without contacting the inner circumferential surface of the heating roller **31** or **31a**. When the fixing belt **203** is used as the fixing member, at least one of the pair of rollers (e.g., the first roller **201** and the second roller **202**), over which the fixing belt **203** is looped, may have a structure similar to the structure of the heating roller **31** or **31a** as described above and the auxiliary heater (e.g., an electric heater and a halogen lamp) may be disposed inside the at least one of the pair of rollers like the heating roller **31** or **31a** as described above.

In the fixing units **11**, **11a**, **11b**, **11c**, **11d**, and **11e**, the fixing members (e.g., the heating rollers **31** and **31a** and the fixing belt **203**) and the pressing members (e.g., the pressure rollers **32** and **204**) are disposed to oppose each other in the horizontal direction. However, the fixing member and the pressing member may be disposed to oppose each other in a vertical direction. For example, the pressing member may be disposed lower than the fixing member to oppose the fixing member. A sheet **S** having a toner image may be conveyed through a nip formed between the fixing member and the pressing member in a substantially horizontal direction.

In the image forming apparatuses **1** and **1b**, a toner image formed on the photoconductor **10** is transferred onto a sheet **S**. However, the toner image formed on the photoconductor **10** may be transferred onto an intermediate transferor (e.g., an intermediate transfer belt) and then transferred from the intermediate transferor to the sheet **S**. The sheet **S** may include paper, cloth, a plastic sheet, and an OHP transparency, which is formed in a sheet-like shape, as long as the sheet **S** may carry the toner image formed thereon and the fixing operation may be performed on the sheet **S** having the toner image.

The present invention has been described above with reference to specific example embodiments. Nonetheless, the present invention is not limited to the details of the example embodiments described above, but various modifications and improvements are possible without departing from the spirit and scope of the present invention. It is therefore to be understood that within the scope of the associated claims, the present invention may be practiced otherwise than as specifically described herein. For example, elements and/or features of different illustrative example embodiments may be combined with each other and/or substituted for each other within the scope of the present patent invention.

What is claimed is:

1. An image forming apparatus, the apparatus comprising: an image forming mechanism to form a toner image on a recording medium according to image data; and a fixing mechanism to fix the toner image on the recording medium, the fixing mechanism including a fixing member to apply heat to the recording medium having the toner image, and a movable external heater configured to move at least one of towards and away from the fixing member to heat the fixing member, the movable external heater being formed in a shape corresponding to a surface of the

31

fixing member, the movable external heater being further configured to at least one of engage and be spaced apart from the fixing member by a distance which is variable according to movement of the movable external heater, the movable external heater being configured to maintain the shape corresponding to the surface of the fixing member regardless as to whether the movable external heater engages or is spaced apart from the fixing member.

2. The image forming apparatus according to claim 1, wherein the distance is reduced to zero so that the movable external heater contacts the fixing member when the fixing member stops and the distance is non-zero so that the movable external heater separates from the fixing member when the fixing member moves.

3. The image forming apparatus according to claim 1, wherein the fixing member includes a convex-like shape portion and the movable external heater includes a concave-like shape portion including an arc corresponding to the convex-like shape portion of the fixing member, and wherein the concave-like shape portion of the movable external heater contacts the convex-like shape portion of the fixing member.

4. The image forming apparatus according to claim 1, wherein the distance is reduced to zero so that the movable external heater contacts the fixing member after the fixing member stops moving and the distance is non-zero so that the movable external heater separates from the fixing member before the fixing member starts moving.

5. The image forming apparatus according to claim 1, wherein the distance is reduced to zero so that the movable external heater contacts the fixing member when the image forming apparatus is powered off.

6. The image forming apparatus according to claim 1, wherein the distance is non-zero such that the movable external heater separates from the fixing member when power is not supplied to the movable external heater.

7. The image forming apparatus according to claim 1, wherein the distance is changed in a sub-scanning direction.

8. The image forming apparatus according to claim 7, wherein the fixing mechanism further includes at least one cam to change the distance, the cam being disposed on a surface of the movable external heater, which does not face the fixing member.

9. The image forming apparatus according to claim 8, wherein the cam includes a non-uniform shape in cross section in the sub-scanning direction.

10. The image forming apparatus according to claim 8, wherein the fixing mechanism further includes at least one belt to pull edges of the external heater to tension the external heater supported by the cam.

11. The image forming apparatus according to claim 1, wherein the fixing mechanism further includes first and second rollers over which the fixing member is looped and a pressing member to apply pressure to the second roller via the fixing member, and wherein the fixing member is formed in a belt-like shape and the movable external heater is disposed along an outer circumferential surface of one of the first and second rollers via the fixing member without contacting the fixing member so as to have a concave-like shape with respect to the outer circumferential surface of the one of the first and second rollers.

32

12. The image forming apparatus according to claim 11, wherein the fixing mechanism further includes one more external heaters disposed along the outer circumferential surface of the other one of the first and second rollers via the fixing member without contacting the fixing member so as to have a concave-like shape with respect to the outer circumferential surface of the other one of the first and second rollers.

13. The image forming apparatus according to claim 1, wherein the movable external heater includes a sheet member and a heat generator buried in the sheet member.

14. The image forming apparatus according to claim 1, wherein the movable external heater is formed in a sheet-like shape having a thickness not greater than about 1 mm.

15. The image forming apparatus according to claim 1, wherein the movable external heater is deformed.

16. The image forming apparatus according to claim 1, wherein the fixing mechanism further includes a heat insulator including a core and a cover and produced by covering the core with the cover in vacuum.

17. The image forming apparatus according to claim 16, wherein the fixing mechanism further includes a unit housing disposed to sandwich the movable external heater together with the fixing member.

18. The image forming apparatus according to claim 17, wherein the movable external heater and the heat insulator are supported by the unit housing.

19. The image forming apparatus of claim 1, wherein the movable external heater is configured to heat the fixing member by conduction and radiation.

20. The image forming apparatus of claim 1, wherein the fixing mechanism further includes a biasing device configured to move the movable external heater towards the fixing member and a moving device configured to move the movable external heater away from the fixing member.

21. The image forming apparatus of claim 20, wherein the biasing device is a spring and the moving device is a cam operatively connected to a motor.

22. The image forming apparatus of claim 21, wherein the cam is oval shaped.

23. A fixing unit for fixing a toner image on a recording medium, the fixing unit comprising:
a fixing member to apply heat to the recording medium having the toner image; and
a movable external heater configured to move at least one of towards and away from the fixing member to heat the fixing member, the movable external heater being formed in a shape corresponding to a surface of the fixing member, the movable external heater being further configured to at least one of engage and be spaced apart from the fixing member by a distance which is variable according to movement of the movable external heater, the movable external heater being configured to maintain the shape corresponding to the surface of the fixing member regardless as to whether the movable external heater engages or is spaced apart from the fixing member.

24. An image forming method comprising:
forming a toner image on a recording medium according to image data; and
fixing the toner image on the recording medium, the fixing step including
disposing an external heater formed in a shape corresponding to a surface of a fixing member as if to engage the fixing member and yet be spaced apart from, the external heater being configured to maintain

33

the shape regardless as to whether the external heater is engaged with the fixing member or spaced apart therefrom,
moving the external heater to be a desired distance apart from the fixing member,

34

heating the fixing member with the external heater, and applying heat to the recording medium having the toner image.

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